

# A Pole Hierarchy for Valley-Monotone Compositions

Resolving a bargraph asymptotic conjecture linked to OEIS A001523,  
with exponential expansions, limit laws, and Lambert- $W$  inversion

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## Abstract

A composition is valley-monotone when the heights of its interior valley plateaux are weakly increasing from left to right. These are the non-decreasing bargraphs of Flórez, Ramírez, and Villamizar (2024). Their area-counting sequence was conjectured to have purely exponential growth, on the basis of a ratio at area 100. We give a self-contained analytic proof of that growth law, exact definitions of its constants, and integer-arithmetic certificates for their numerical values. The growth constant is determined directly by the generating function of the classical weakly unimodal composition sequence OEIS A001523.

The proof reveals a stronger structure. A decomposition by minimum part produces infinitely many genuine, simple positive poles, increasing to one, whose coefficient contributions alternate in sign. Consequently the area generating function is not D-finite and its coefficients are not P-recursive. We prove a finite-pole expansion to every prescribed radius below one, a central limit theorem for the number of bottom-level valleys, and an exponentially tight limiting terminal component. The accumulating real poles admit an all-orders inverse-logarithmic expansion expressed through Lambert  $W$ , with an error beyond every inverse-logarithmic order. We also give exponential-sector inversion and the necessary integer threshold qualifications. The claims are conventional mathematical proofs, accompanied by reproducible checks, not Lean formalizations or independently peer-reviewed results.

[write] **Added 5 October 2026, batch 98.** Since batch 98 the report has two Parts. Part I, summarized above, is the original report. Part II (Section 11 onward) adds manuscript 04 of batch 98: it answers Part I's extreme-height question with a lattice-Gumbel law for the maximum part, introduces a valley-weighted model with a first-order transition at an explicit critical weight, and proves critical-window, Dirichlet and Dirichlet-mixture laws, an exact height-tail identity with an expansion of the height-cutoff pole, and inverses of these expansions. Its labels carry the prefix `vmb:cw:`.

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# 1 The question, the OEIS connection, and the scope of the result

## 1.1 A published asymptotic conjecture

Let  $d(n)$  count nonempty compositions of  $n$  whose valley heights are weakly increasing. A valley is a maximal constant run with larger parts on both sides; endpoint runs are not valleys. For example,  $(3, 1, 2, 2, 4, 2, 3)$  has valley heights 1, 2, whereas  $(3, 2, 3, 1, 2)$  has heights 2, 1 and is excluded. The area of the corresponding bargraph is the sum of its column heights. The sequence begins

$$1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1023, 2042, 4071, 8106, 16121, \dots$$

The convention throughout is  $d(0) = 0$ .

Conjecture 2.3 of Flórez, Ramírez, and Villamizar [1] proposes an exponential equivalent for this sequence. The displayed numerical estimates in that conjecture are

$$\lambda_{100} = 1.9764206780767492582\dots, \quad C_{100} = 0.6039015148966519\dots \quad (1.1)$$

Here the growth estimate is the finite ratio  $d(100)/d(99)$ , not an exact definition of the limiting exponential base. We establish

$$d(n) = C\lambda^n(1 + O(\theta^n)), \quad 0 < \theta < 1, \quad (1.2)$$

with

$$\begin{aligned} \lambda &= 1.97642067794583573531465245217332\dots, \\ C &= 0.60390151906189910496030721536293\dots \end{aligned}$$

Thus the proposed *form* is correct, while some displayed numerical digits must be replaced. Interpreting a finite decimal base literally in an asymptotic equivalent would be incorrect: an arbitrarily small fixed error in that base becomes exponentially significant as  $n$  grows.

## 1.2 What is, and is not, being claimed about OEIS

Let

$$A(q) = \sum_{n \geq 0} a(n)q^n, \quad a(n) = \text{OEIS A001523}(n). \quad (1.3)$$

This is the classical sequence of weakly unimodal compositions, with  $a(0) = 1$ . Its leading stretched-exponential asymptotic is already known; the OEIS entry credits Auluck (1951) and links the 2024 bargraph paper [2, 1]. We do *not* claim to solve a previously open leading-asymptotic problem for A001523 itself.

Instead, the classical sequence is the exact input to a new pole analysis of the associated valley-monotone class. In particular, the reciprocal growth constant  $\rho = \lambda^{-1}$  is the unique solution in  $(0, 1)$  of

$$\boxed{\rho(1 - \rho)A(\rho) = 1 + \rho.} \quad (1.4)$$

No OEIS identifier for  $d(n)$  was verified in the searches for this report, so none is invented. The connection to OEIS is both explicit and structural, rather than an accidental match of a few terms.

### 1.3 Relationship to ProveIt and novelty boundary

The repository’s **Analysis/Transseries** project separates asymptotic scales, flatness, and inverse expansions from the Fabius function development [4]. The present article fits that program, but does not require any unproved theorem from the repository. A repository search for **bargraphs** and **A001523** returned no matching files. This was a search-level overlap check, not an audit of every archive in the repository. The inspected repository tree had SHA **29aca108ed25d714f31b1316512777fbbdc8e006**.

The generating-function decomposition is due to the 2024 paper; it is reproved below in minimum-part notation. The contributions developed here are the analytic resolution of the conjectured exponential form, the full positive-pole hierarchy and non-D-finiteness, the pole accumulation expansion, and the probabilistic and inverse consequences. Targeted searches located no earlier proof of this particular conjecture or these stronger conclusions. That is a bounded literature-search statement, not a guarantee of priority. General meromorphic coefficient extraction and analytic inversion are standard tools [3].

**[write] Added 2 October 2026, batch 75.** This report is a single manuscript, printed in full: manuscript 04 of batch 75 of ProveIt’s incoming-reports intake, delivered as the archive **OEIS\_Valley\_Monotone\_Bargraphs.zip** in commit **4b874cea0** and placed in commit **6ea60e367** in the research-report collection, among the enumerative-combinatorics reports (**SetTheory/Cardinals/docs/reports**), not in the **Analysis/Transseries** project named above. The identifier called a “tree SHA” above is the ProveIt *commit* **29aca108e** (1 October 2026), an ancestor of the placement commit; it is this manuscript’s pin. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No OEIS number for the sequence  $d(n)$  was verified, and the writing step assigns none; **A001523** is the input sequence, not the target. The writing step changed the labels (every label now carries the prefix **vmb:**), renamed Theorem 7.3 (delivered as “Pole accumulation transseries”), set the bibliography ragged-right, and added these **[write]** notes and one reference [5]; no other statement, proof or number of the manuscript was altered. The Lambert- $W$  and rounding steps of Sections 7 and 8 are instances of the transseries-and-inversion volume [5], as the notes there say; no transseries in that volume’s sense is constructed here.

**[write] Added 5 October 2026, batch 98.** The note above describes Part I, which was a single manuscript printed in full. Since batch 98 the report also has a Part II, built from a second manuscript (manuscript 04 of batch 98, pin **70ab31a1e**, placed in commit **9353a7171**); Section 11.1 records its provenance and Section 11.3 its notation, which differs from Part I’s in places listed there.

## 2 A decomposition by minimum part

### 2.1 Classes and notation

For an integer  $j \geq 1$ , let  $\mathcal{U}_j$  be the nonempty weakly unimodal compositions whose parts are at least  $j$ , and let  $\mathcal{D}_j$  be the nonempty valley-monotone compositions with that same lower bound. Write

$$U_j(q) = \sum_{\alpha \in \mathcal{U}_j} q^{|\alpha|}, \quad D_j(q) = \sum_{\alpha \in \mathcal{D}_j} q^{|\alpha|}, \quad d_j(n) = [q^n]D_j(q). \quad (2.1)$$

Here  $|\alpha|$  is the sum of the parts. In particular,  $U_1 = A - 1$ ,  $D_1(q) = \sum_{n \geq 1} d(n)q^n$ . Define

$$H_j(q) = \frac{q^j}{1 - q^j}, \quad K_j(q) = H_j(q)U_{j+1}(q), \quad Q_j(q) = 1 - q^j - q^j U_{j+1}(q). \quad (2.2)$$

Thus

$$Q_j(q) = (1 - q^j)(1 - K_j(q)). \quad (2.3)$$

The series  $H_j$  counts a nonempty run of parts equal to  $j$ .

**Lemma 2.1** (Unimodal decomposition). *As formal power series,*

$$U_j = H_j + \frac{U_{j+1}}{(1 - q^j)^2}. \quad (2.4)$$

*Equivalently,*

$$U_j(q) = \sum_{h \geq j} \frac{q^h}{1 - q^h} \prod_{r=j}^{h-1} (1 - q^r)^{-2}. \quad (2.5)$$

*Proof.* An object either consists only of parts  $j$ , or has a nonempty middle object with all parts greater than  $j$ . In the second case, unimodality allows parts  $j$  only in the initial and final runs. Those runs may be empty and are chosen independently, giving  $(1 - q^j)^{-2}$ . This proves the recurrence. Iteration classifies an object by its maximum part  $h$  and gives the sum. For each coefficient of fixed degree, only finitely many values of  $h$  contribute, so the formal iteration is legitimate.  $\square$

**Lemma 2.2** (Valley-monotone decomposition). *As formal power series,*

$$D_j = H_j + \frac{D_{j+1}}{(1 - q^j)^2(1 - K_j)} = H_j + \frac{D_{j+1}}{(1 - q^j)Q_j}. \quad (2.6)$$

*Proof.* The all- $j$  object contributes  $H_j$ . Otherwise split the composition into its maximal blocks of parts greater than  $j$ , with runs of  $j$ 's between them and possibly at the two ends. Each internal separating run is a valley of height  $j$ .

Every block except the last must be unimodal. Indeed, an internal valley in an earlier block would have height greater than  $j$  and would be followed by a separating valley of height  $j$ , violating the required order. A finite sequence without an internal valley is weakly unimodal: a downward change followed later by an upward change necessarily creates an intervening valley. Conversely, unimodal earlier blocks create no additional valleys and the last block may be any member of  $\mathcal{D}_{j+1}$ .

The unique decomposition is therefore

$$(\text{initial } j\text{-run}) (\mathcal{U}_{j+1} \text{ nonempty } j\text{-run})^* \mathcal{D}_{j+1} (\text{final } j\text{-run}).$$

The two endpoint factors contribute  $(1 - q^j)^{-2}$ ; the repeated unit has generating function  $K_j$ . This proves the formula, including uniqueness and the absence of overcounting.  $\square$

**Corollary 2.3** (Iterated series). *For  $j \geq 1$ ,*

$$D_j(q) = \sum_{h \geq j} H_h(q) \prod_{r=j}^{h-1} \frac{1}{(1 - q^r)Q_r(q)}. \quad (2.7)$$

*In particular,*

$$D_1(q) = \sum_{h \geq 1} \frac{q^h}{\prod_{r=1}^h (1 - q^r) \prod_{r=1}^{h-1} Q_r(q)}. \quad (2.8)$$

This agrees with the area specialization in [1], where the paper's  $B_j$  is our  $U_{j+1}$ . The boundary condition at large  $j$  is coefficientwise disappearance:  $U_j, D_j$  have no terms below degree  $j$ . It uniquely determines both recurrences.

## 2.2 The renewal kernel in terms of A001523

From (2.4) at  $j = 1$ ,

$$U_2 = (1 - q)^2 U_1 - q(1 - q).$$

Consequently,

$$K_1(q) = q(1 - q)A(q) - q = \sum_{n \geq 3} (a(n - 1) - a(n - 2))q^n. \quad (2.9)$$

Thus the differences of a classical OEIS sequence count the repeated units. This identity will also provide the probability distribution underlying the limit theorem.

## 3 Analytic continuation and the positive-pole hierarchy

### 3.1 Normal convergence inside the unit disk

**Lemma 3.1** (Uniform estimates). *Fix  $0 < R < 1$ . Uniformly for  $|q| \leq R$ , all the functions  $U_j$  are analytic and*

$$|U_j(q)| \leq B_R R^j, \quad |K_j(q)| = O_R(R^{2j+1}), \quad Q_j(q) = 1 + O_R(R^j), \quad (3.1)$$

where  $B_R$  is finite and independent of  $j$ .

*Proof.* The infinite product  $P_R = \prod_{r \geq 1} (1 - R^r)^{-2}$  is finite, since the sum of the absolute logarithms converges. Each term in (2.5) is bounded by  $P_R R^h / (1 - R)$ . Summing for  $h \geq j$  gives  $B_R = P_R / (1 - R)^2$ , and local uniform convergence proves analyticity. The other estimates follow from (2.2).  $\square$

**Theorem 3.2** (Meromorphic continuation). *Every  $D_j$  extends to a single-valued meromorphic function on  $|q| < 1$ . Its possible poles there are among the zeros of  $Q_r$ ,  $r \geq j$ . On each compact subdisk there are only finitely many possible poles.*

*Proof.* For sufficiently large  $r$ , uniformly on a chosen compact subdisk,  $Q_r$  stays away from zero and

$$\frac{1}{(1 - q^r)Q_r(q)} = 1 + O_R(R^r).$$

The corresponding tail products in (2.7) are uniformly bounded, because  $\sum_r R^r$  converges. The summands are therefore  $O_R(R^h)$  after separating finitely many initial denominator factors. The tail sum converges normally off those factors' zeros. Multiplying by the finite product of initial  $Q_r$ 's makes the sum analytic locally. Each initial  $Q_r$  is analytic and not identically zero, so its zeros are isolated and finite in a compact subdisk. These observations prove all the assertions.  $\square$

### 3.2 Distinguished real roots

**Lemma 3.3.** *For every  $j \geq 1$  there is a unique  $\rho_j \in (0, 1)$  with  $K_j(\rho_j) = 1$ , equivalently  $Q_j(\rho_j) = 0$ . Moreover,*

$$0 < \rho_1 < \rho_2 < \cdots < 1, \quad \rho_j \longrightarrow 1, \quad Q'_j(\rho_j) < 0. \quad (3.2)$$

*The only zero of  $Q_j$  on  $|q| \leq \rho_j$  is  $q = \rho_j$ .*

*Proof.* On  $(0, 1)$ ,  $K_j$  is strictly increasing, starts at zero, and tends to infinity. The last claim follows already from the subclass consisting of one part greater than  $j$ , together with the nonempty separating run. Thus  $\rho_j$  exists uniquely and its derivative is positive. Equation (2.3) then gives  $Q'_j(\rho_j) = -(1 - \rho_j^j)K'_j(\rho_j) < 0$ .

For positive  $q < 1$ , both  $H_j > H_{j+1}$  and  $U_{j+1} > U_{j+2}$ . Hence  $K_j > K_{j+1}$ , proving strict ordering of their unit crossings. The estimates in Lemma 3.1 show that  $K_j(q) \rightarrow 0$  for every fixed  $q < 1$ ; this forces  $\rho_j \rightarrow 1$ .

If  $|q| < \rho_j$ , positivity gives  $|K_j(q)| \leq K_j(|q|) < 1$ , excluding zeros. On the boundary, equality with  $K_j(q) = 1$  in the triangle inequality would force all nonzero summands of  $K_j(q)$  to be nonnegative real. Its support contains both  $2j + 1$  and  $2j + 2$ : use a separating part  $j$  and a singleton of height  $j + 1$  or  $j + 2$ . Their consecutive exponents force  $q/\rho_j = 1$ . This proves boundary uniqueness.  $\square$

**Theorem 3.4** (Genuine simple poles and alternating amplitudes). *For every  $k \geq j$ ,  $D_j$  has a genuine simple pole at  $\rho_k$ . In particular, write the principal part of  $D_1$  there as*

$$\frac{C_k}{1 - q/\rho_k}.$$

Then

$$C_k = \frac{D_{k+1}(\rho_k)}{(1 - \rho_k^k)(-\rho_k Q'_k(\rho_k))} \prod_{r=1}^{k-1} \frac{1}{(1 - \rho_k^r)Q_r(\rho_k)}, \quad \text{sgn}(C_k) = (-1)^{k-1}. \quad (3.3)$$

These are exactly the positive real poles inside the unit disk.

*Proof.* By Lemma 3.3,  $D_{k+1}$  is analytic at  $\rho_k$ : all of its possible pole factors have indices greater than  $k$  and larger minimum zero modulus. Its defining positive series converges there and has positive value. Applying (2.6) at level  $k$  therefore produces a genuine simple pole, of amplitude

$$\frac{D_{k+1}(\rho_k)}{(1 - \rho_k^k)(-\rho_k Q'_k(\rho_k))} > 0.$$

For every  $r < k$ , the recurrence at level  $r$  adds an analytic term and multiplies the inherited pole by a finite nonzero factor  $((1 - q^r)Q_r(q))^{-1}$ . Since  $Q_r(\rho_k) < 0$ , each such factor changes its sign. This proves the formula, noncancellation, and sign. Theorem 3.2 and uniqueness of each positive zero exclude any other positive poles.  $\square$

**Corollary 3.5** (Non-D-finiteness and non-P-recursiveness). *None of the functions  $D_j$  is D-finite over  $\mathbb{C}(q)$ . In particular  $d(n)$  satisfies no eventual nonzero linear recurrence with polynomial coefficients in  $n$ .*

*Proof.* A function satisfying a linear differential equation with polynomial coefficients can have finite-plane singularities only at the finitely many zeros of its leading coefficient. At every other point, standard local existence for a linear analytic differential equation continues its analytic branches holomorphically. The infinitely many poles  $\rho_k$ ,  $k \geq j$ , contradict this requirement.

For completeness, a polynomial-coefficient recurrence for the Taylor coefficients, after multiplication by powers of  $q$  and summation, yields a linear differential equation for the generating function, using  $q d/dq$  to encode multiplication by  $n$ . Any finite initial discrepancy is polynomial and may be annihilated by further differentiation. Thus an eventual recurrence would also imply D-finiteness, which has just been excluded.  $\square$

## 4 The exponential law and certified constants

**Theorem 4.1** (Leading asymptotics at every fixed minimum part). *For each fixed  $j \geq 1$ , there is  $R_j > \rho_j$ , with  $R_j < 1$ , such that*

$$d_j(n) = c_j \rho_j^{-n} + O_j(R_j^{-n}), \quad c_j = \frac{D_{j+1}(\rho_j)}{(1 - \rho_j^j)^2 \rho_j K'_j(\rho_j)} > 0. \quad (4.1)$$

For  $j = 1$ ,  $c_1 = C_1 = C$  and  $\rho_1 = \rho$ , so this proves (1.2) and the intended form of Conjecture 2.3 of [1].

*Proof.* Lemma 3.3 and Theorem 3.4 give a unique simple pole on the circle of convergence. Local finiteness of all possible poles permits a slightly larger closed disk containing no others. Subtract the principal part  $c_j/(1 - q/\rho_j)$ . The difference is analytic on a neighborhood of that disk. Cauchy's coefficient bound on its boundary gives the stated error. The formula for  $c_j$  follows from (2.3). Equation (2.9) gives (1.4).  $\square$

There are no nonzero inverse-power corrections multiplying the leading exponential: for every fixed  $M$ ,

$$\frac{d(n)}{C\lambda^n} = 1 + o(n^{-M}). \quad (4.2)$$

The missing information is instead in genuinely smaller exponential sectors. This is why an inverse-power fit to the leading term would conceal the analytic structure.

### 4.1 Exact definitions and integer-arithmetic enclosures

The constants are defined by

$$K_1(\rho) = 1, \quad \lambda = \rho^{-1}, \quad C = \frac{D_2(\rho)}{(1 - \rho)^2 \rho K'_1(\rho)}. \quad (4.3)$$

The accompanying `verify.py` proves the following enclosures with outward-rounded rational intervals, using integers only:

$$\begin{aligned} 0.50596515770080666537292224451833314155847753935777 &< \rho, \\ \rho &< 0.50596515770080666537292224451833314155847753935778. \end{aligned} \quad (4.4)$$

$$1.976420677945835735314652452172 < \lambda < 1.976420677945835735314652452175, \quad (4.5)$$

$$0.603901519061899104960307215362 < C < 0.603901519061899104960307215364. \quad (4.6)$$

The narrower machine-produced intervals are retained in `verification.json`. These bounds certify, in particular, the numerical distinction between the limit and the finite-area estimates.

**Proposition 4.2** (Tail bounds used in the certificate). *For  $0 < q < 1$ , put  $S_J(q) = q^{J+1}/(1-q)$ . If  $S_J(q) < 1$ , then*

$$0 \leq U_{J+1}(q), D_{J+1}(q) \leq \frac{S_J(q)}{1 - S_J(q)}, \quad 0 \leq U'_{J+1}(q) \leq \frac{S'_J(q)}{(1 - S_J(q))^2}. \quad (4.7)$$

*Proof.*  $S_J$  counts a single allowed part. All nonempty ordered sequences of such parts have generating function  $S_J/(1 - S_J)$ , and contain both restricted classes as subclasses. Coefficientwise domination proves the first two inequalities and, after differentiating nonnegative coefficients, the derivative bound.  $\square$

The program uses  $J = 250$  and endpoints in  $10^{-90}\mathbb{Z}$ , propagates these tail intervals down to level 2, and tests the sign of  $Q_1$  at the two rational endpoints in (4.4). To enclose  $C$ , it uses the equivalent expression

$$C = \frac{D_2(\rho)}{\rho(1 - \rho)\{1 + U_2(\rho) + \rho U'_2(\rho)\}}. \quad (4.8)$$

Every division checks that its denominator interval excludes zero. No unbounded numerical tail is discarded.

## 5 Exponential expansions to arbitrary subunit radius

**Theorem 5.1** (Finite-pole expansion). *Choose  $0 < R < 1$  such that  $D_1$  has no pole on  $|q| = R$ , and let  $\mathcal{P}_R$  be its finite set of actual poles inside that circle. Write the principal part at  $\zeta \in \mathcal{P}_R$  in the form*

$$\sum_{\ell=1}^{m_\zeta} \frac{c_{\zeta,\ell}}{(1 - q/\zeta)^\ell}.$$

Then

$$d(n) = \sum_{\zeta \in \mathcal{P}_R} \sum_{\ell=1}^{m_\zeta} c_{\zeta,\ell} \binom{n+\ell-1}{\ell-1} \zeta^{-n} + O_R(R^{-n}). \quad (5.1)$$

*Proof.* Subtract all the displayed principal parts. The result is analytic in a neighborhood of the closed disk, so its coefficients are  $O_R(R^{-n})$  by Cauchy's formula. The coefficient of each subtracted rational term is the displayed binomial coefficient times  $\zeta^{-n}$ .  $\square$

For positive real poles,  $m_{\rho_k} = 1$  and  $c_{\rho_k,1} = C_k$ . Thus these sectors have no polynomial prefactors. Nonreal conjugate pairs may yield exponentially small oscillations; multiplicities or coincidences among complex zeros may yield polynomial factors. Formula (5.1) makes no unsupported simplicity assumption for those zeros.

The numerical positive-real hierarchy begins as follows. The table's ordering is by *positive real pole*, not by modulus of all complex poles. Apart from the first row, these decimal entries are numerical checks rather than rational certificates.

| $k$ | $\rho_k$          | $\rho_k^{-1}$     | $C_k$              |
|-----|-------------------|-------------------|--------------------|
| 1   | 0.505965157700807 | 1.976420677945836 | 0.603901519061899  |
| 2   | 0.628520593778803 | 1.591037763755333 | -0.434266071916591 |
| 3   | 0.694915251028261 | 1.439024396889127 | 0.097127120171435  |
| 4   | 0.738032724347822 | 1.354953468877238 | -0.008189474598318 |
| 5   | 0.768802380990748 | 1.300724379536014 | 0.000290088602322  |
| 6   | 0.792097830048713 | 1.262470318771736 | -0.000004651415177 |

*Remark 5.2* (Meaning of “all orders”). For every fixed exponential cutoff  $R < 1$ , only finitely many sectors are required and the remainder is proved. This does not assert that the infinite sum of all pole contributions converges, that the unit circle is a natural boundary, or that the positive-real poles alone constitute the complete expansion. In particular, we have not certified that  $\rho_2$  is the second-smallest modulus among *all* poles. An all-orders algebraic expansion around the dominant term is trivial here; the finite-pole theorem is the substantive exponential refinement.

## 6 A renewal limit law and a tight terminal component

### 6.1 Bottom-level valleys

Choose uniformly from the  $d(n)$  objects of area  $n$ , and let  $V_n$  be the number of valleys of height 1. Introduce a variable  $u$  marking these valleys. The decomposition in Lemma 2.2 gives

$$D(q, u) = H_1(q) + \frac{B(q)}{1 - uK(q)}, \quad B(q) = \frac{D_2(q)}{(1 - q)^2}, \quad K(q) = K_1(q). \quad (6.1)$$

The repeated unit contributes exactly one height-1 valley. Define a positive integer random variable  $Z$  by

$$\mathbb{P}(Z = s) = [q^s]K(q) \rho^s. \quad (6.2)$$

This is a probability distribution because  $K(\rho) = 1$ . It has exponential moments near zero and includes both  $s = 3$  and  $s = 4$  in its support. Put

$$M = \mathbb{E}Z = \rho K'(\rho), \quad V = \text{Var } Z = \rho^2 K''(\rho) + \rho K'(\rho) - M^2 > 0. \quad (6.3)$$

**Theorem 6.1** (Gaussian bottom-valley count). *As  $n \rightarrow \infty$ ,*

$$\frac{V_n - \mu n}{\sqrt{\sigma^2 n}} \implies \mathcal{N}(0, 1), \quad \mu = \frac{1}{M}, \quad \sigma^2 = \frac{V}{M^3}. \quad (6.4)$$

Moreover  $\mathbb{E}V_n = \mu n + O(1)$  and  $\text{Var } V_n = \sigma^2 n + O(1)$ . Numerical evaluation gives

$$\mu = 0.137248756066383542\dots, \quad \sigma^2 = 0.0368978521690088765\dots$$

*Proof.* The implicit function theorem gives an analytic root  $r(u)$  near  $u = 1$ , satisfying  $uK(r(u)) = 1$  and  $r(1) = \rho$ . Choose a closed disk slightly larger than  $\rho$  in which (6.1) at  $u = 1$  has only its simple pole. Compactness and analytic perturbation preserve exactly this one nearby pole for complex  $u$  sufficiently close to 1. Subtracting it uniformly gives

$$\mathbb{E}[u^{V_n}] = \frac{C(u)}{C(1)} \left( \frac{\rho}{r(u)} \right)^n (1 + O(\vartheta^n)), \quad C(u) = \frac{B(r(u))}{ur(u)K'(r(u))}, \quad (6.5)$$

for some  $\vartheta < 1$ , locally uniformly in  $u$ .

Set  $t(s) = \log r(e^s)$ . Taking logarithms of the root equation gives  $\log K(e^{t(s)}) = -s$ . The first two derivatives of  $\log K(e^t)$  at  $t = \log \rho$  are respectively  $M$  and  $V$ . Therefore

$$t'(0) = -M^{-1}, \quad t''(0) = -VM^{-3}.$$

Taking logarithms in (6.5) and differentiating twice proves the mean and variance statements. Substituting  $s = iz/\sqrt{\sigma^2 n}$ , subtracting the centering term, and using the analytic Taylor expansion gives a limiting characteristic function  $\exp(-z^2/2)$ . This proves the asserted weak convergence.  $\square$

### 6.2 Why almost all valleys are at height one

In the nonflat decomposition, let  $T_n \in \mathcal{D}_2$  be the final block with parts greater than one, and let  $L_n, R_n \geq 0$  be the lengths of the two endpoint runs of ones. The all-ones composition has probability  $1/d(n)$ . On this exceptional object, define  $(L_n, R_n) = (0, 0)$  and let  $T_n = \dagger$  be a cemetery object of area zero. This convention is asymptotically irrelevant.

**Theorem 6.2** (Boltzmann terminal limit). *The triple  $(L_n, T_n, R_n)$  converges in distribution to independent components  $(L, T, R)$ , where*

$$\mathbb{P}(L = \ell) = \mathbb{P}(R = \ell) = (1 - \rho)\rho^\ell, \quad \mathbb{P}(T = \tau) = \frac{\rho^{|\tau|}}{D_2(\rho)} \quad (\tau \in \mathcal{D}_2). \quad (6.6)$$

*The sizes of these components have uniformly bounded exponential moments for all sufficiently large  $n$ . The number of valleys above height one is therefore  $O_{\mathbb{P}}(1)$ , and the total number of valleys has the same central limit theorem as  $V_n$ .*

*Proof.* For a fixed triple  $(\ell, \tau, r)$ , its contribution is  $q^{\ell+|\tau|+r}/(1-K(q))$ . Comparing its dominant residue with that of (6.1) at  $u = 1$  gives limiting probability

$$\frac{\rho^{\ell+|\tau|+r}}{B(\rho)} = (1 - \rho)^2 \rho^{\ell+r} \frac{\rho^{|\tau|}}{D_2(\rho)},$$

which factors as stated and sums to one.

To justify tightness and exponential moments, mark endpoint length by  $v$  and terminal area by  $w$ . The numerator becomes  $D_2(wq)/(1-vq)^2$ . Since  $D_2$  is analytic past  $\rho$ , this numerator remains analytic on a common larger disk when real  $v, w > 1$  are sufficiently close to 1. The same residue argument then bounds the marked expectation uniformly in  $n$ . All valleys above height one lie in  $T_n$ , and their number is at most its area. Their contribution divided by  $\sqrt{n}$  tends to zero in probability, so Slutsky's theorem transfers the central limit law to the total valley count.  $\square$

The limiting terminal area has mean

$$\mathbb{E}|T| = \frac{\rho D_2'(\rho)}{D_2(\rho)} = 6.2446300331882976716 \dots$$

This decimal is numerical, not one of the certified intervals above. The same arguments work at every fixed level  $j$ , with  $K_j$ ,  $D_{j+1}$ , and runs of parts  $j$ . As an additional consequence of Theorem 4.1,

$$\mathbb{P}(\text{all parts are at least } 2) \sim \frac{c_2}{c_1} \left( \frac{\rho_1}{\rho_2} \right)^n. \quad (6.7)$$

Here  $c_2 > 0$  is the *dominant amplitude* of  $D_2$ , not the negative amplitude  $C_2$  of the second positive pole of  $D_1$ .

## 7 How the real poles accumulate at one

### 7.1 An exact product comparison

For  $0 < q < 1$ , define

$$P_{j+1}(q) = \prod_{h \geq j+1} (1 - q^h)^{-2}. \quad (7.1)$$

**Lemma 7.1.** *If  $y = q^j$ , then*

$$\frac{1 - yq}{2 - yq} (P_{j+1} - 1) \leq U_{j+1} \leq \frac{1}{2} (P_{j+1} - 1). \quad (7.2)$$

*Proof.* Let  $A_h = \prod_{r=j+1}^h (1 - q^r)^{-2}$ , with  $A_j = 1$ . The term of  $U_{j+1}$  with maximum part  $h$  is

$$\frac{q^h}{1 - q^h} A_{h-1} = \frac{1 - q^h}{2 - q^h} (A_h - A_{h-1}).$$

The multiplicative factor lies between  $(1 - yq)/(2 - yq)$  and  $1/2$ . Sum the positive telescoping increments and pass to the limit.  $\square$

Put

$$t_j = -\log \rho_j, \quad x_j = jt_j, \quad y_j = e^{-x_j} = \rho_j^j, \quad c = \log 2. \quad (7.3)$$

At the pole, the defining equation is  $U_{j+1}(\rho_j) = (1 - y_j)/y_j$ .

**Lemma 7.2** (Pole-scale equation). *As  $j \rightarrow \infty$ ,  $x_j \rightarrow \infty$ ,  $x_j \sim \log j$ , and*

$$\frac{2je^{-x_j}}{x_j} = x_j + c + O\left(\frac{x_j^3}{j}\right). \quad (7.4)$$

*Proof.* We know  $t_j \rightarrow 0$ . We first show  $y_j \rightarrow 0$ . Otherwise some subsequence has  $y_j \geq \delta > 0$ . The singleton subclass gives  $U_{j+1}(\rho_j) \geq \rho_j^{j+1}/(1 - \rho_j) \rightarrow \infty$ , whereas  $(1 - y_j)/y_j$  is bounded, a contradiction. Thus  $x_j \rightarrow \infty$ .

At the root, (7.2) gives

$$\log P_{j+1}(\rho_j) = x_j + c + O(y_j). \quad (7.5)$$

On the other hand, expanding the product logarithm absolutely gives, with  $t = t_j$ ,  $y = y_j$ ,

$$\begin{aligned} \log P_{j+1}(e^{-t}) &= 2 \sum_{k \geq 1} \frac{y^k}{k(e^{kt} - 1)} \\ &= \frac{2y}{t} + O\left(y + \frac{y^2}{t}\right). \end{aligned} \quad (7.6)$$

For the last estimate use  $(e^t - 1)^{-1} = t^{-1} + O(1)$  for  $k = 1$ , and  $(e^{kt} - 1)^{-1} \leq (kt)^{-1}$  for  $k \geq 2$ .

Since  $y \rightarrow 0$ , the error is  $o(y/t)$ , while the right side of (7.5) tends to infinity. It follows that  $2y/t \sim x$ , or  $2je^{-x} \sim x^2$ . Taking logarithms gives  $x + 2 \log x = \log(2j) + o(1)$ , hence  $x \sim \log j$  and  $y \asymp x^2/j$ . Substituting this estimate back into (7.6) and (7.5) gives (7.4).  $\square$

## 7.2 A Lambert– $W$ expansion to all logarithmic orders

**Theorem 7.3** (Pole accumulation: a Lambert– $W$  expansion). *Let  $X_j > 0$  be the unique positive solution of*

$$e^{X_j} X_j (X_j + c) = 2j, \quad c = \log 2, \quad (7.7)$$

*and set*

$$w_j = 2W\left(\sqrt{j/2}\right), \quad (7.8)$$

*where  $W$  is the positive real Lambert function. Then*

$$x_j = X_j + O\left(\frac{(\log j)^2}{j}\right). \quad (7.9)$$

*For every fixed  $m \geq 1$ ,*

$$x_j = w_j + \sum_{k=1}^m \frac{\alpha_k(c)}{w_j^k} + O_m(w_j^{-m-1}) + O\left(\frac{w_j^2}{j}\right), \quad (7.10)$$

where

$$\begin{aligned}\alpha_1(c) &= -c, \\ \alpha_2(c) &= \frac{c(c+4)}{2}, \\ \alpha_3(c) &= -\frac{c(c^2+6c+12)}{3}, \\ \alpha_4(c) &= \frac{c(3c+8)(c^2+6c+12)}{12}.\end{aligned}\tag{7.11}$$

The full series for  $X_j - w_j$  is convergent for sufficiently large  $w_j$ ; the difference between  $x_j$  and this logarithmic sector is smaller than every fixed inverse power of  $\log j$ .

*Proof.* Let  $F_j(x) = 2je^{-x}/x - x - c$ . Its derivative is negative. Lemma 7.2 gives  $F_j(x_j) = O(x_j^3/j)$ , while  $F_j(X_j) = 0$ . First,  $F'_j \leq -1$  gives  $|x_j - X_j| = O(x_j^3/j) = o(1)$ . Thus throughout the interval between these two points,  $|F'_j| \asymp x_j$ , because  $2je^{-x_j}/x_j \sim x_j$ . A second application of the mean value theorem improves the difference to  $O(x_j^2/j)$ , proving (7.9).

The definition of  $w = w_j$  is equivalent to  $e^w w^2 = 2j$ . Write  $X = w + \varepsilon$ ,  $z = 1/w$ . Taking logarithms of (7.7) and cancelling the equation for  $w$  gives

$$\varepsilon + 2 \log(1 + z\varepsilon) + \log\left(1 + \frac{cz}{1 + z\varepsilon}\right) = 0.\tag{7.12}$$

This is analytic near  $(z, \varepsilon) = (0, 0)$ , with derivative 1 with respect to  $\varepsilon$ . The analytic implicit function theorem supplies a convergent series  $\varepsilon = \sum_{k \geq 1} \alpha_k(c) z^k$ . Coefficient comparison gives (7.11), and gives every further coefficient recursively: substitute the already computed terms, and negate the coefficient of  $z^k$  to obtain  $\alpha_k$ . The new coefficient occurs only in the first, linear term at that order. Combining this convergent expansion with (7.9) proves all claims.  $\square$

In particular,

$$1 - \rho_j \sim \frac{\log j}{j}, \quad x_j = \log j - 2 \log \log j + \log 2 + o(1).\tag{7.13}$$

Higher expansions for  $\rho_j = e^{-x_j/j}$  and the growth bases  $\rho_j^{-1} = e^{x_j/j}$  now follow by ordinary exponential composition, with the error propagated explicitly. The theorem is a large- $j$  statement about the pole locations; it does not by itself make the fixed- $j$  coefficient estimate uniform when  $j$  grows with  $n$ .

[write] **Added 2 October 2026, batch 75: an instance of repository results.** The manuscript titled Theorem 7.3 “Pole accumulation transseries”; it is a single-scale Lambert- $W$  expansion of the pole positions, and the theorem has been renamed accordingly. Taking logarithms, (7.7) reads  $X + 2 \log X = \log(2j) - \log(1 + c/X)$ . Its dominant block  $X + 2 \log X = \log(2j)$  is solved exactly by the Lambert core `p0:thm:lambert-core` of the transseries-and-inversion volume [5] (there  $a = 1$ ,  $b = 2 > 0$ , principal branch), which gives  $w_j = 2W(\sqrt{j/2})$ . The correction equation (7.12) is that volume’s reversion about an arbitrary core, `p0:thm:core-reversion`, with  $\Lambda = 1$ ,  $h = 0$  and small parameter  $t = z = 1/w_j$ ; its triangular recurrence is the coefficient recursion described in the proof, and `code/expansion.py` implements it. The leading equation of Corollary 7.4, in the variable  $x + c$ , is another instance of that core (there  $a = b = 1$ ), which gives  $x + c = W(4/t)$ . These are instances of that apparatus, and no novelty is claimed for the method. What is specific to this report is the pole-scale equation (Lemma 7.2) and the error terms in (7.9) and (7.10).

### 7.3 Inverting the real-pole counting function

**Corollary 7.4.** *Let  $N_+(r) = \#\{j \geq 1 : \rho_j \leq r\}$ , and put  $t = -\log r$ . Then, as  $r \uparrow 1$ ,*

$$N_+(r) = \frac{W(4/t) - \log 2}{t} + O\left(\log \frac{1}{t}\right). \quad (7.14)$$

*Proof.* At  $r = \rho_j$ , rewrite (7.4) as

$$\frac{2e^{-x}}{t} = x + c + O(tx^2), \quad x = jt.$$

The leading equation has solution  $x = W(4/t) - c$ , since  $e^c = 2$ . Its derivative has order  $x$ , so the same two-stage mean-value argument gives an error  $O(tx)$  in  $x$ , or  $O(x)$  in  $j$ . Also  $x \asymp \log(1/t)$ . Between consecutive real poles, monotonicity sandwiches the counting function between the two adjacent values. The leading expression in (7.14) is decreasing in  $t$ , so the same error estimate holds throughout the gaps.  $\square$

## 8 Inverting the area-counting asymptotics

### 8.1 Finite-sector real interpolants

Choose a cutoff  $R > \rho$  as in Theorem 5.1. The finite principal-part expansion can be interpolated for real  $x$  as

$$F_R(x) = Ce^{Lx}(1 + E_R(x)), \quad L = \log \lambda > 0. \quad (8.1)$$

For a nonreal conjugate pair, use its real oscillatory expression; for a negative real pole use  $r^{-x} \cos(\pi x)$  in place of  $(-r)^{-n}$ , with the same polynomial prefactor. This prescription agrees with every integer coefficient contribution and is real analytic. All nonleading terms and each fixed derivative are bounded by a polynomial times  $e^{-\delta x}$ , for some  $\delta > 0$ , because there are only finitely many poles in the chosen expansion. Consequently  $F_R$  is strictly increasing for sufficiently large  $x$ . This is a specified asymptotic interpolant, not a claim that a discrete sequence has a unique natural real interpolation.

**Theorem 8.1** (Inverse-sector expansion). *Let  $x_R(Y)$  solve  $F_R(x) = Y$  on its eventually increasing branch, and put  $x_0 = L^{-1} \log(Y/C)$ . Set  $\Phi(x) = -L^{-1} \log(1 + E_R(x))$ . Then*

$$x_R(Y) = x_0 + \sum_{k \geq 1} \frac{1}{k!} \left( \frac{d}{dx} \right)^{k-1} \Phi(x)^k \Big|_{x=x_0}. \quad (8.2)$$

*For each fixed cutoff and sufficiently large  $Y$ , this is a convergent small-correction expansion. In particular,*

$$x_R(Y) = x_0 - \frac{E_R(x_0)}{L} + \frac{E_R(x_0)^2}{2L} + \frac{E_R(x_0)E'_R(x_0)}{L^2} + O\left(P(x_0)e^{-3\delta x_0}\right) \quad (8.3)$$

*for a suitable fixed polynomial  $P$ .*

*Proof.* Writing  $x = x_0 + h$  and taking logarithms in (8.1) gives the exact equation  $h = \Phi(x_0 + h)$ . On a fixed small complex neighborhood of  $h = 0$ ,  $\Phi$  and its derivative tend uniformly to zero as  $x_0 \rightarrow \infty$ . Analytic Lagrange inversion applies; it may also be obtained by inserting a parameter  $s$  into  $h = s\Phi(x_0 + h)$ , expanding at  $s = 0$ , and using the uniform smallness to continue the series past  $s = 1$ . This proves (8.2). Expanding  $\Phi = -E_R/L + E_R^2/(2L) + O(E_R^3)$ , the terms with  $k = 1, 2$  give (8.3); all remaining terms have at least three exponentially small factors, with polynomial derivative bounds.  $\square$

For one simple real sector  $E_R(x) = be^{-\Delta x}$ , the first terms are especially transparent:

$$x_R(Y) = x_0 - \frac{b}{L} \left(\frac{Y}{C}\right)^{-\Delta/L} + \left(\frac{1}{2L} - \frac{\Delta}{L^2}\right) b^2 \left(\frac{Y}{C}\right)^{-2\Delta/L} + \dots \quad (8.4)$$

For the  $\rho_k$  sector,  $b = C_k/C$  and  $\Delta = \log(\rho_k/\rho)$ . With several poles, products create mixed exponent sums. One must retain any other sector of comparable or larger size before interpreting a displayed correction as the next term of an ordered asymptotic expansion.

## 8.2 Integer thresholds and the rounding obstruction

**Proposition 8.2.** *The sequence  $d(n)$  is strictly increasing for  $n \geq 1$ . Define  $N(Y) = \min\{n \geq 1 : d(n) \geq Y\}$ . For each fixed cutoff  $R$ , there exists  $K_R > 0$  such that for all sufficiently large  $Y$ ,*

$$\lceil x_R(Y) - K_R Y^{-\eta_R} \rceil \leq N(Y) \leq \lceil x_R(Y) + K_R Y^{-\eta_R} \rceil, \quad \eta_R = \frac{\log(R/\rho)}{\log \lambda} > 0. \quad (8.5)$$

*Proof.* Append a part 1 to any valid composition. No new valley is created, since the new step is nonascending; this is an injection from area  $n$  to area  $n+1$ . The singleton  $(n+1)$  is not in its image, proving strict increase.

At integers, Theorem 5.1 gives  $d(n) = F_R(n) + O_R(R^{-n})$ . Near  $x_R(Y)$ ,  $F'_R(x) \asymp Y$ , so the corresponding uncertainty in the crossing location is bounded by a constant times  $R^{-x_R}/Y = O(Y^{-\eta_R})$ . One may make this precise first within a fixed distance of the crossing, using the mean value theorem; outside that distance the strict leading exponential separates the two sides. All integer points below the left shift have count below  $Y$ , and all those above the right shift have count at least  $Y$ . Taking ceilings yields the bounds.  $\square$

A single unconditional identity  $N(Y) = \lceil x_R(Y) \rceil$  would be unjustified when  $Y$  lies arbitrarily close to an actual integer threshold. Formula (8.5) records the missing rounding information rather than hiding it in a real-variable error term.

[write] **Added 2 October 2026, batch 75: an instance of repository results.** Proposition 8.2 and the remark after it are the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [5]: part (1) gives  $N(Y) = \lceil \nu(Y) \rceil$  for any increasing interpolation  $\nu$  of the sequence, and part (2), the separation condition with  $\eta = K_R Y^{-\eta_R}$ , says that the two ceilings in (8.5) coincide, and then equal  $N(Y)$ , unless  $x_R(Y)$  lies within  $\eta$  of an integer. The generic arithmetic of those two parts is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, statements about an arbitrary monotone interpolation that give no formal status to this proposition. The series (8.2) of Theorem 8.1 is Lagrange's reversion formula (in formal form, `p0:thm:LB` of the same volume). No novelty is claimed for these methods.

## 9 Reproducible computations and proof checks

### 9.1 Independent exact enumeration

The package includes two logically independent finite checks. First, a triangular coefficient algorithm implements (2.4)–(2.6) using integers and formal-series long division. Initializing  $U_{N+1} = D_{N+1} = 0$  gives exactly all coefficients through degree  $N$ ; omitted objects have area larger than  $N$ . Second, exhaustive enumeration generates every composition through area 17, collapses consecutive equal parts, extracts interior local minima, and checks their order directly.

It agrees with the coefficient algorithm. Lower bounds on the parts equal to 2 and 3 are also checked independently through area 13.

The nonempty  $U_1$  coefficients agree with the displayed A001523 values through  $n = 20$ ; its zero-degree coefficient differs only because  $A = 1 + U_1$ . The exact computation also gives

$$d(100) = 233832578565556052089825216845.$$

The program retains all coefficients through 200 in `verification.json`. Finite checks support the implementation; they are not substitutes for the asymptotic proofs.

## 9.2 Convergence of the finite ratios

Let  $\varepsilon_n = d(n)/(C\lambda^n) - 1$ . High-precision numerical comparison gives:

| $n$ | $d(n)/d(n-1)$          | $\varepsilon_n$             |
|-----|------------------------|-----------------------------|
| 10  | 2.00000000000000000000 | $-6.7798174 \cdot 10^{-2}$  |
| 20  | 1.9808570502327331233  | $-9.1891465 \cdot 10^{-3}$  |
| 30  | 1.9769260944011981878  | $-1.0612096 \cdot 10^{-3}$  |
| 50  | 1.9764273761888468482  | $-1.4002503 \cdot 10^{-5}$  |
| 75  | 1.9764207075861918763  | $-6.1918466 \cdot 10^{-8}$  |
| 100 | 1.9764206780767492582  | $-2.7346103 \cdot 10^{-10}$ |
| 150 | 1.9764206779458382883  | $-5.3327913 \cdot 10^{-15}$ |
| 200 | 1.9764206779458357354  | $-1.0399431 \cdot 10^{-19}$ |

The finite-area quantities reproduce both estimates in (1.1):

$$\frac{d(100)}{d(99)} = 1.976420678076749258223519681232\dots,$$

$$\frac{d(100)}{(d(100)/d(99))^{100}} = 0.603901514896651955795442836948\dots$$

This identifies the source of the discrepancy exactly; it is not a change in the combinatorial definition or in the area indexing.

## 9.3 Files and execution

`verify.py` requires only the Python standard library. It checks exact enumeration, agreement with the supplied classical initial terms, and the rigorous rational intervals. Run

```
python verify.py
```

The separate `numerics.py` requires `mpmath` and produces high-precision exploratory values for poles, residues, and limit-law constants. Its output explicitly distinguishes those numerical values from certified enclosures. A small optional `expansion.py` uses `sympy` to regenerate the  $\alpha_k(c)$  coefficients from (7.12). The article is self-contained and can be compiled with

```
pdflatex article.tex
pdflatex article.tex
```

No source font files, downloaded papers, or repository files are bundled.

[write] **Added 2 October 2026, batch 75.** In the repository the programs live in `code/` and their recorded outputs in `data/`. Each program writes its output file into its own directory (`verify.py` writes `verification.json`, `numerics.py` writes `numerical_results.json` and imports `verify`, `expansion.py` writes `expansion_coefficients.json`), so the report’s README runs them on a scratch copy. The delivered build script and PDF are not shipped; the README gives the build command.

## 10 Further research questions and topics

### 10.1 The nonreal pole spectrum

Determine the smallest-modulus nonreal poles, and certify their multiplicities and ordering relative to  $\rho_2, \rho_3, \dots$ . An effective route is to combine the explicit tail bounds with argument-principle or Rouché certificates on rationally described contours. This would turn the abstract exponential cutoff in Theorem 5.1 into a numerically explicit first several sectors, and could yield sharp eventual error signs or ratio monotonicity. These latter conclusions have *not* been inferred from the numerical table.

### 10.2 Is the unit circle a natural boundary?

The real poles prove non-D-finiteness and obstruct meromorphic extension through 1. They do not prove a natural boundary at every point of the unit circle. A proof would need to locate noncancelling singularities approaching a dense set of boundary points, or otherwise preclude local continuation there. The factors  $Q_j$  and the root-of-unity structure of the unimodal products give a concrete starting point.

### 10.3 The next sector of pole accumulation

The estimate  $x_j - X_j = O((\log j)^2/j)$  is beyond all logarithmic orders. Determine its leading coefficient and a full expansion in powers of  $1/j$ , with coefficient functions of  $w_j$ . The proof suggests a systematic calculation: retain the  $k = 2$  term in (7.6), Euler–Maclaurin corrections to  $(e^t - 1)^{-1}$ , and a sharper weighted telescoping expansion in Lemma 7.1. The resulting two-scale transseries should improve the pole-counting error beyond  $O(\log(1/t))$ .

### 10.4 Joint limits and extreme heights

The renewal representation can mark the number of columns, the number of ones, peak heights, and the maximum height inside a repeated unit. A joint normal law for additive statistics should follow after proving nondegeneracy of the appropriate covariance matrix. Maximum height is not additive: determine whether its logarithmic scale exhibits a lattice Gumbel law and quantify any periodic phase. No extreme-value law is claimed in this report.

[write] **Added 5 October 2026, batch 98: answered in part.** The extreme-height half of this question is answered in Part II: Theorem 17.1 proves  $\mathbb{P}(H_n \leq h) = \exp(-\alpha n \rho^h)(1 + O(\log n/n))$  for the maximum part  $H_n$  in the window  $n \rho^h \asymp 1$ , with the explicit intensity  $\alpha = \rho^2 P_1(\rho)/M$ ; (17.6) gives the lattice laws along phase subsequences, so there is no single continuous Gumbel limit; and Theorem 17.2 shows that the maximum is asymptotically independent of the Gaussian bottom-valley count. The sentence “No extreme-value law is claimed in this report” describes Part I. The first half, a joint normal law for additive statistics (columns, ones, peak heights) with a nondegenerate covariance matrix, is not addressed by Part II and stays open (Subsection 21.8, item 7).

## 10.5 Moving lower bounds and a crossover problem

Theorem 4.1 holds for fixed  $j$ , while Theorem 7.3 describes large  $j$ . Determine a uniform estimate for  $d_j(n)$  when  $j = j(n)$  grows. Since the pole gap shrinks as  $j$  increases, a one-pole approximation need not remain uniform. Locating the scale at which multiple poles interact would connect the fixed-level exponential regime with sparse-composition regimes.

## 10.6 Formalization and effective algorithms

A useful ProveIt contribution would formalize the finite combinatorial decomposition first, including the plateau convention, then develop a general locally finite pole-product lemma. The exact interval certificate is suitable for a small independently checked arithmetic kernel. A separate algorithmic question is whether the exact coefficients can be obtained substantially faster than triangular long division, using the rapidly increasing minimum degrees of the kernels and fast truncated multiplication. Non-P-recursiveness rules out a fixed polynomial recurrence, not fast coefficient algorithms.

## Part II. Critical windows and extreme heights

*Sections 1–10 and Appendices A–B form the original report (Part I) of 1 October 2026, unchanged except for dated notes. Part II was added on 5 October 2026 in batch 98 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the extreme-height question of Part I (Section 10, “Joint limits and extreme heights”), introduces a valley-weighted model with a first-order transition, and develops its critical windows. Section 11 records the provenance and fixes the notation of the addition; its appendices follow Part I’s.*

### 11 Provenance, scope and notation of Part II

[write] **Added 5 October 2026, batch 98.** This section was written in the batch-98 write, except Subsection 11.4, which prints the manuscript’s own first section with the changes listed there.

#### 11.1 Provenance

Part II is manuscript 04 of batch 98, the archive `OEIS_Bargraph_Critical_Windows_and_Extremes.zip`, which arrived in commit `2172df76a` and was placed in this report by commit `9353a7171`. Its article is titled *Critical Windows and Extreme Heights in Valley-Monotone Compositions*, with the subtitle “An OEIS A001523-linked continuation of ProveIt: phase coexistence, Dirichlet laws, and invertible exponential expansions” (24 pages, dated October 4, 2026); its author line reads “Research manuscript prepared for Vladimir Reshetnikov”. AI assistance is not stated in the manuscript. Its pinned repository commit is `70ab31a1ec77852d411bc8ca1fc81d86d0c9985e` (4 October 2026), and the article blob it read, `f37593705acecfdc8dc158e9079ae36887e96609`, is the text of Part I as committed in the batch-75 write (`9891e1f5e`); Part I had not changed between that pin and this write, so nothing in the manuscript is stale with respect to it. The manuscript’s source notes are shipped as `02-critical-windows-SOURCE_NOTES.md`; its programs and recorded outputs are shipped with the prefix `02-critical-windows-` in `code/` and `data/` (Section 20).

The manuscript cites OEIS A001523 (accessed October 4, 2026), the 2024 bargraph paper [1], whose bibliographic record and OEIS linkage it checked but whose full publisher text *was not accessible* in its session, Flajolet and Sedgewick [3] (through the author-maintained page <https://algo.inria.fr/flajolet/Publications/books.html>), and Part I at the pinned commit, which it treats as a research draft, “not ... an independently reviewed publication”. In Part II these references are printed as the entries [2], [1], [3] of Part I’s bibliography, with the manuscript’s access notes kept in this paragraph, and its citations of Part I are printed as cross-references.

#### 11.2 What Part II proves, and how it relates to Part I

Part I proves the exponential law, the positive-pole hierarchy, the bottom-valley central limit theorem and a Boltzmann terminal limit for the uniform model. Part II keeps the same objects and adds:

- the *lattice-Gumbel law* of the maximum part,  $\mathbb{P}(H_n \leq h) = \exp(-\alpha n \rho^h)(1 + O(\log n/n))$  in the window  $n \rho^h \asymp 1$ , its lattice phase, and the asymptotic independence of  $H_n$  from the

Gaussian bottom-valley count (Theorems 17.1 and 17.2);

- a weighted model with three regimes in the bottom-valley weight  $u$ , the exact critical weight  $u_c = \rho_2(1 - \rho_2)(1 - \rho_2^2)/(1 + \rho_2^4)$ , a uniform  $1/n$  critical window and its all-orders expansion (Theorems 13.1, 14.1, Proposition 14.2);
- Dirichlet area laws at simultaneous critical weights on finitely many levels (Theorem 15.2) and the Dirichlet-mixture law of the maximum part there (Theorem 18.1);
- an exact  $q$ -product tail identity and the expansion of the height-cutoff pole to every exponential order (Lemma 16.1, Theorem 16.2);
- inverses of these expansions with rounding qualifications (Section 19).

It also re-proves parts of Part I so as to be independent of them. Table 1 lists every overlap. Where Part II's statement coincides with Part I's, Part I's statement stands and Part II's text is a second route; the generalizations are new.

| Part II                                       | Relation to Part I or to the repository                                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 12.1, (12.5)                            | second route to Lemmas 2.1, 2.2 and (2.5) (the decomposition is originally [1]); the weighted form (12.4) at every level is new and generalizes (6.1), which marks level 1 only |
| Lemma 12.2                                    | second route to Theorem 3.2, Lemma 3.3 and Theorem 4.1 (same amplitudes $c_j$ )                                                                                                 |
| Theorem 13.1                                  | its case $u = 1$ (supercritical) is Theorem 6.1, and the tightness of the final block there is part of Theorem 6.2; the regimes $u \neq 1$ are new                              |
| Proposition 19.3                              | its monotonicity argument is that of Proposition 8.2; its ceilings are an instance of <code>p0:thm:staircase</code> [5]                                                         |
| (19.4)–(19.5), (19.7)–(19.8),<br>Theorem 19.2 | instances of <code>p0:thm:lambert-core</code> , <code>p0:thm:core-reversion</code> and <code>p0:thm:LB</code> [5] (note in Section 19); the constants are new                   |
| everything else                               | new in this report: Theorems 13.1 ( $u \neq 1$ ), 14.1, 15.2, 16.2, 17.1, 17.2, 18.1, 19.2, Proposition 14.2, Lemmas 15.1, 16.1, Corollaries 13.2, 15.3                         |

Table 1: Overlaps of Part II with Part I and with the repository.

**Answered and open questions of Part I.** Part I's question “Joint limits and extreme heights” (Section 10) asked two things. Its second half, whether the maximum height has a lattice Gumbel law and what its periodic phase is, is answered by Theorem 17.1 and (17.6), and Theorem 17.2 adds the joint law of the bottom-valley count and the maximum. Its first half, a joint normal law for additive statistics (columns, ones, peak heights) with a nondegenerate covariance matrix, is *not* addressed by Part II and stays open. Part I's questions on the natural boundary and on moving lower bounds are restated, not answered, in Section 21.

**What the write proved.** Five statements of the manuscript are supported only by a sketch, an analogy or a one-line remark. The write supplies proofs of them, each marked [write] and printed with its proof: the tilted multicritical window (Proposition 15.4), the fixed-parameter phase-chamber classification (Proposition 15.5), the conditional intensity of the critical maximum and its dependence on the area fractions (Proposition 18.2), the multicritical rounding and critical quantile bounds (Proposition 19.4), and the monotonicity of the interpolated cutoff shift (Lemma 19.1). These proofs have not been independently reviewed. Section 21 lists them with the manuscript's open questions.

### 11.3 Notation

Part II uses Part I's names for every shared object. The manuscript uses several letters that Part I uses for other objects, and overloads some internally ( $a, b, c, C, D, T, p, x_0, a_r$ ). Table 2 fixes the convention; every renaming is listed, and no normalization is changed. The macros `\Pp` and `\coef` of the manuscript are printed with Part I's `\Prob` and `\coeff`; the manuscript's unused macros `\c0` and `\tr` are not carried over. Short identifications with Part I set as `[write: ...]` were added in writing on 5 October 2026, like the `[write]` notes.

Table 2: Notation of Part II against the manuscript and Part I.

| Notion                                    | Part II                                           | Manuscript; remark                                                                                                 |
|-------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| second positive pole                      | $\rho_2$                                          | $R$ (Part I's $R, R_j$ are radii)                                                                                  |
| pole at the $m$ -level critical point     | $\rho_{m+1}$                                      | $R_m$ ; also “ $R$ ” in its Section 8                                                                              |
| radius beyond the dominant pole           | $R, R_j$ (as in Theorem 4.1)                      | $T, T_j$ (Part I's $T$ is the limiting terminal block)                                                             |
| moments of $Z$ at $\rho = \rho_1$         | $M = M_\rho, V = V_\rho$ of (6.3)                 | $M_\rho, V_\rho$                                                                                                   |
| critical moments at $\rho_2$              | $M_c, V_c, J_c, D_c = V_c/(2M_c^3)$               | $M, V, J, D$ (Part I's $M, V$ are at $\rho$ ; Part I's $D_j$ are series)                                           |
| window amplitude                          | $d_c$                                             | $d$ (not Part I's $d(n)$ )                                                                                         |
| first window correction                   | $G_1(s)$                                          | $C_0(s)$ (equal by Proposition 14.2)                                                                               |
| leading coefficient at $m$ levels         | $\kappa^{(m)}$ , with $\kappa^{(1)} = \kappa$     | $C_m$ (Part I's $C_k$ are amplitudes at $\rho_k$ )                                                                 |
| critical counting polynomial              | $\Pi_m(x) = \kappa^{(m)}x^m(1 + \pi_1/x + \dots)$ | $P_m, p_k$                                                                                                         |
| area in level- $j$ repeated units         | $\mathcal{A}_{j,n}$                               | $C_{j,n}$                                                                                                          |
| area of the final block                   | $ T_n $ (Part I's block $T_n$ )                   | $T_n$                                                                                                              |
| Dirichlet coordinates                     | $\xi_0, \dots, \xi_m, \xi$                        | $x_0, \dots, x_m, \mathbf{x}$ (Part I's $x_j = jt_j$ ; the manuscript's $x_0$ is also a leading inverse)           |
| area marks                                | $v_j$                                             | $t_j$ (Part I's $t_j = -\log \rho_j$ )                                                                             |
| product $\prod_{r \geq j} (1 - q^r)^{-2}$ | $P_j(q), P_1(q)$ (Section 7)                      | $\mathcal{P}_j, \mathcal{P}$ (Part I's $\mathcal{P}_R$ is a set of poles)                                          |
| cutoff polynomials                        | $\varpi_{r-1}(h)$                                 | $Q_{r-1}(h)$ (Part I's $Q_j$ is a series)                                                                          |
| order- $r$ cutoff coefficients            | $A_r(q)$                                          | $a_r(q)$ in a proof, $A_r(q)$ in Appendix B (the $a_j(q)$ of (16.8) are level- $j$ coefficients; $a_1 = A_1$ only) |
| product coefficients                      | $\gamma_\ell(q)$                                  | $c_\ell(q)$ (Part I's $c_j$ are amplitudes)                                                                        |
| capped amplitude                          | $\hat{c}_h$                                       | $c_h$                                                                                                              |
| level-2 numerator, Laurent constant       | $B_2 = E_2 D_3, c_{2,0}$                          | $A_2, d_{20}$                                                                                                      |
| nearby-pole numerator                     | $\Xi_n, \Xi_n(\rho_{m+1}) \rightarrow a_*$        | $A_n, a$                                                                                                           |
| subcritical ratio                         | $\bar{u} = u/u_c$                                 | $a$                                                                                                                |
| height-window bounds                      | $\tau_- \leq n\rho^h \leq \tau_+$                 | $a \leq n\rho^h \leq b$                                                                                            |
| inverse shift                             | $a_c = \kappa_0/\kappa$                           | $a$                                                                                                                |

| Notion                           | Part II                                      | Manuscript; remark (continued)                                    |
|----------------------------------|----------------------------------------------|-------------------------------------------------------------------|
| height centres                   | $\beta_n, \beta_n^{\text{crit}}, \beta_p(n)$ | $b_n, b_n^{\text{crit}}, b_p(n)$ (not the Laurent $b, b_0$ )      |
| standard normal limit            | $\chi$                                       | $Z$ (Part I's $Z$ is the unit size)                               |
| multicritical inverse correction | $\varphi$                                    | $h$ (a height elsewhere)                                          |
| number of minimal roots          | $\nu$                                        | $p$                                                               |
| log-series variables in (19.5)   | $z_c, \ell_1, \ell_2$                        | $Z, L_1, L_2$ (its $L_m$ at $m = 1$ is $L_c$ )                    |
| rounding constant                | $C_*$                                        | $C$ (Part I's $C$ is the growth amplitude)                        |
| <hr/>                            |                                              |                                                                   |
| <i>Kept, with a warning</i>      |                                              |                                                                   |
| marked function                  | $F(q, u)$                                    | $= D(q, u)$ of (6.1); not the interpolant $F_R$                   |
| $\log(1/\rho)$                   | $L$                                          | $= \log \lambda$ , Part I's $L$                                   |
| Gumbel intensities               | $\alpha, \alpha_j^*$                         | not the coefficients $\alpha_k(c)$ of (7.11)                      |
| weighted denominator             | $Q_{j,u}$                                    | $Q_{j,1} = (1 - q^j)Q_j$ with Part I's $Q_j$                      |
| lattice phase                    | $\theta$                                     | not the ratio $\theta$ of (1.2)                                   |
| cutoff shift                     | $\delta_h, \delta(h)$                        | not the decay rate $\delta$ of (8.1)                              |
| run kernel, $(1 - q^j)^{-2}$     | $H_j, E_j$                                   | $H_j$ as in Part I; $H_n$ is the maximum part; $E_j$ is not $E_R$ |
| simplex                          | $\Delta_m$                                   | not the gap $\Delta$ of (8.4)                                     |
| dominant amplitude of $D_j$      | $c_j$                                        | as in Theorem 4.1                                                 |

## 11.4 The manuscript's research question, sources and contribution boundary

[write] **Added 5 October 2026, batch 98.** This subsection is the manuscript's first section; its three subsections are printed as unnumbered headings. Its citations of Part I are printed as cross-references, and nothing else is changed.

### The OEIS connection is structural

OEIS A001523 counts weakly unimodal compositions, or stacks. Its classical asymptotics are already known, so the present article does not present them as an open problem. Write

$$A(q) = \sum_{n \geq 0} \text{A001523}(n) q^n. \quad (11.1)$$

The entry gives, among other formulas,

$$A(q) = 1 + \sum_{h \geq 1} \frac{q^h}{1 - q^h} \prod_{r=1}^{h-1} (1 - q^r)^{-2}. \quad (11.2)$$

It also links the 2024 paper of Flórez, Ramírez, and Villamizar on non-decreasing bargraphs [2, 1]. The term “non-decreasing” refers to the heights of successive interior valley plateaux, not to all columns. We use the more explicit description *valley-monotone*.

The repository report *A Pole Hierarchy for Valley-Monotone Compositions* (Part I) rederives the minimum-part decomposition and proves its first counting and probability consequences. Its final research section asks whether maximum height has a lattice-Gumbel law and how joint statistics behave. Those are specific, documented questions addressed here. The target bargraph counting sequence is not identified with A001523: A001523 supplies the unimodal blocks. No unverified OEIS number for the target sequence is assigned.

## What is inherited and what is developed here

The symbolic decomposition, the unweighted dominant pole, the positive-pole hierarchy, and the bottom-valley central limit theorem are inherited from Part I; the decomposition originates in [1]. We give the short proofs needed to make the present article independent of unproved repository assertions. Meromorphic coefficient extraction, quasi-powers arguments, Lagrange inversion, and the appearance of simplex laws from products of simple poles are standard methods of analytic combinatorics [3].

The present continuation develops the following model-specific conclusions: the exact maximum-height intensity and its lattice phase; a weighted first-order transition with a uniform critical density; its finite-size scaling function and correction; a finite hierarchy of simultaneous critical weights with Dirichlet area sharing; the resulting mixed extreme-height law; and an explicit all-orders expansion and inversion for the moving height-cutoff pole. The maximum-height theorem answers the explicit question in Part I. The weighted and multicritical formulations introduce and solve further questions rather than claim to settle a previously advertised OEIS conjecture.

## Audit and limitations

The repository source inspected was

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/valley-monotone-bargraphs/
article.tex
```

at the snapshot identified by commit 70ab31a1ec77852d411bc8ca1fc81d86d0c9985e; the fetched article blob was f37593705acecfdc8dc158e9079ae36887e96609. The relevant definitions, analytic results, and final research questions were read, not inferred from filenames. Targeted searches for the published bargraph title and for maximum-height, phase-transition, and Dirichlet refinements did not identify an earlier matching result. The publisher's complete article was not accessible in this session; its bibliographic record and the OEIS linkage were accessible. Consequently this is a bounded source audit, not an exhaustive novelty certificate. All decimals introduced below are numerical diagnostics unless explicitly described otherwise.

## 12 Minimum-part decomposition and analytic foundations

### 12.1 Objects and generating functions

A composition is a nonempty finite sequence of positive integers. Its area is the sum of its parts, and its height is its largest part. Collapse every maximal constant run to a single entry. An interior entry smaller than both neighbors represents one valley plateau. A composition is valley-monotone when the resulting valley heights are weakly increasing.

Let  $\mathcal{U}_j$  and  $\mathcal{D}_j$  be, respectively, the nonempty unimodal and valley-monotone compositions with all parts at least  $j$ . Their ordinary generating functions are  $U_j(q)$  and  $D_j(q)$ . Thus  $U_1 = A - 1$ . We use

$$H_j(q) = \frac{q^j}{1 - q^j}, \quad E_j(q) = (1 - q^j)^{-2}, \quad K_j(q) = H_j(q)U_{j+1}(q). \quad (12.1)$$

The symbol  $H_j$  counts a nonempty run at level  $j$ ; it is not the random maximum height, which will be denoted  $H_n$ .

**Lemma 12.1** (Unique decomposition). *As formal power series,*

$$U_j = H_j + E_j U_{j+1}, \quad (12.2)$$

$$D_j = H_j + \frac{E_j D_{j+1}}{1 - K_j}. \quad (12.3)$$

*If each valley of height  $j$  receives weight  $u_j$ , the corresponding weighted function satisfies*

$$D_j(q; u_j, u_{j+1}, \dots) = H_j(q) + \frac{E_j(q) D_{j+1}(q; u_{j+1}, \dots)}{1 - u_j K_j(q)}. \quad (12.4)$$

*Only finitely many  $u_j$  need differ from one.*

*Proof.* A unimodal object either is an all- $j$  run or has a nonempty central object with larger parts, flanked by two possibly empty  $j$ -runs. This proves (12.2).

For a valley-monotone object that is not all  $j$ , split at the runs of  $j$ . Each nonfinal block of larger parts must be unimodal: an interior valley there would have height larger than  $j$  and would precede a valley of height  $j$ . Conversely, choosing unimodal nonfinal blocks and a valley-monotone final block creates no forbidden descent among valley heights. The unique form is

$$(\text{initial } j\text{-run}) (\mathcal{U}_{j+1} \text{ nonempty } j\text{-run})^* \mathcal{D}_{j+1} (\text{final } j\text{-run}).$$

Every repeated block contributes exactly one height- $j$  valley. Its kernel is  $K_j$ , or  $u_j K_j$  after marking. This proves both remaining formulas.  $\square$

[write] **Added 5 October 2026, batch 98.** (12.2) and (12.3) are (2.4) and (2.6) of Lemmas 2.1 and 2.2, and this proof is a second route to them. The weighted form (12.4) at every level  $j$  is new; at  $j = 1$  with  $u_2 = u_3 = \dots = 1$  it is (6.1).

Iterating (12.2) gives

$$U_j(q) = \sum_{h \geq j} \frac{q^h}{1 - q^h} \prod_{r=j}^{h-1} (1 - q^r)^{-2}. \quad (12.5)$$

The unweighted target sequence  $d(n) = [q^n] D_1(q)$ , with  $d(0) = 0$ , begins

$$1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1023, 2042, 4071, 8106, 16121, \dots$$

The equality with  $2^{n-1}$  through area ten is only an initial coincidence.

## 12.2 A local analytic foundation, with no global conjectures

**Lemma 12.2** (Dominant poles). *For every fixed  $j$ ,  $U_j$  is analytic on  $|q| < 1$ . There is a unique  $\rho_j \in (0, 1)$  such that  $K_j(\rho_j) = 1$ , and*

$$\rho_j < \rho_{j+1}, \quad \rho_j \longrightarrow 1.$$

*The function  $D_j$  is meromorphic on  $|q| < 1$ . Its unique singularity on  $|q| \leq \rho_j$  is a simple pole at  $\rho_j$ . For some  $R_j > \rho_j$ ,*

$$[q^n] D_j = c_j \rho_j^{-n} + O(R_j^{-n}), \quad c_j = \frac{E_j(\rho_j) D_{j+1}(\rho_j)}{M_j(\rho_j)} > 0, \quad (12.6)$$

where

$$M_j(q) = \frac{q K_j'(q)}{K_j(q)}, \quad V_j(q) = q \frac{d}{dq} M_j(q). \quad (12.7)$$

*Proof.* For  $|q| \leq r < 1$ , the product in (12.5) is uniformly bounded by  $\prod_{k \geq j} (1 - r^k)^{-2} < \infty$ , and the remaining summand is  $O(r^h)$ . This gives local normal convergence. For real  $q$ ,  $K_j(q)$  increases strictly from zero to infinity. Both  $H_j > H_{j+1}$  and  $U_{j+1} > U_{j+2}$ , so  $K_j > K_{j+1}$  and the roots are strictly ordered.

For large  $j$  and fixed  $r < 1$ , all compositions with allowed parts at least  $j$  are bounded coefficientwise by  $S_j/(1 - S_j)$ , where  $S_j(q) = q^j/(1 - q)$  and  $S_j(r) < 1$ . Thus  $K_j = O(r^{2j})$  locally as  $j \rightarrow \infty$ , which proves  $\rho_j \rightarrow 1$ . Iteration of (12.3) gives a locally normally convergent sum away from the locally finite zeros of  $1 - K_j$ : its multiplier is  $E_j/(1 - K_j) = 1 + O(r^j)$  and its additive term is  $H_j = O(r^j)$ . This proves meromorphic continuation.

For  $k > j$ ,  $K_k(\rho_j) < 1$ , so the higher-level factors are analytic on  $|q| \leq \rho_j$ . The support of  $K_j$  contains the consecutive degrees  $2j + 1$  and  $2j + 2$ . Equality in the triangle inequality in  $K_j(q) = 1$ , with  $|q| = \rho_j$ , therefore forces  $q = \rho_j$ . The pole is simple because  $K'_j(\rho_j) > 0$ , and its numerator is positive. Subtract  $c_j/(1 - q/\rho_j)$  and apply Cauchy's coefficient bound on a slightly larger circle containing no other singularity.  $\square$

[write] **Added 5 October 2026, batch 98.** Lemma 12.2 restates Theorem 3.2, Lemma 3.3 and Theorem 4.1, and its proof is a condensed second route to them. Its  $\rho_j$  and  $c_j$  are Part I's: since  $K_j(\rho_j) = 1$ ,  $M_j(\rho_j) = \rho_j K'_j(\rho_j)$  and the two formulas for  $c_j$  agree. The  $S_j$  in the proof is Part I's  $S_j$  of Proposition 4.2 with  $J = j - 1$ .

For real  $q \in (0, 1)$ ,  $M_j(q)$  and  $V_j(q)$  are the mean and variance of the positive integer distribution

$$\mathbb{P}(Z_j = s) = \frac{[q^s]K_j(q)q^s}{K_j(q)}. \quad (12.8)$$

In particular  $V_j(q) > 0$ . No natural-boundary assertion or classification of nonreal poles is required in this paper. [write: at  $j = 1$ ,  $q = \rho$  this is Part I's  $Z$  of (6.2), and  $M_1(\rho) = M$ ,  $V_1(\rho) = V$  are Part I's moments (6.3).]

### 13 A bottom-valley phase transition

Let  $V_n$  count the valleys of height one. Give an area- $n$  object weight  $u^{V_n}$ , with  $u > 0$ , and normalize these weights to obtain a probability measure  $\mathbb{P}_{n,u}$ . Define

$$F(q, u) = H_1(q) + \frac{B(q)}{1 - uK(q)}, \quad K = K_1, \quad B = E_1 D_2, \quad f_n(u) = [q^n]F(q, u). \quad (13.1)$$

[write: this  $F$  is Part I's  $D(q, u)$  of (6.1), with the same  $B$  and  $K$ .] Put

$$u_c = \frac{1}{K(\rho_2)}, \quad M_c = M_1(\rho_2), \quad V_c = V_1(\rho_2), \quad J_c = \frac{u_c \rho_2^2 K''(\rho_2)}{2}. \quad (13.2)$$

At  $\rho_2$  expand

$$B(q) = \frac{b}{1 - q/\rho_2} + b_0 + O(1 - q/\rho_2), \quad b > 0. \quad (13.3)$$

All these constants are defined analytically, not by their displayed decimals.

**Theorem 13.1** (Three regimes). *For fixed  $u > 0$  the following statements hold.*

**Subcritical regime**,  $u < u_c$ . Set  $\bar{u} = u/u_c$ . There is  $R > \rho_2$  such that

$$f_n(u) = \frac{b}{1 - \bar{u}} \rho_2^{-n} + O(R^{-n}), \quad V_n \implies \text{Geom}_0(1 - \bar{u}), \quad (13.4)$$

where the limiting probability at  $k \geq 0$  is  $(1 - \bar{u})\bar{u}^k$ . The final  $\mathcal{D}_2$  block has area  $n - O_{\mathbb{P}}(1)$ .

**Supercritical regime,**  $u > u_c$ . Let  $r = r(u) < \rho_2$  solve  $uK(r) = 1$ . For some  $R > r$ ,

$$f_n(u) = \frac{B(r)}{M_1(r)} r^{-n} + O(R^{-n}), \quad (13.5)$$

$$\frac{V_n - n/M_1(r)}{\sqrt{nV_1(r)/M_1(r)^3}} \implies \mathcal{N}(0, 1). \quad (13.6)$$

The final block has tight area, with an exponentially decaying limiting tail.

**Critical regime,**  $u = u_c$ . There is  $R > \rho_2$  such that

$$f_n(u_c) = \rho_2^{-n}(\kappa n + \kappa_0) + O(R^{-n}), \quad (13.7)$$

where

$$\kappa = \frac{b}{M_c}, \quad \kappa_0 = \frac{b}{M_c} + \frac{b_0}{M_c} + \frac{bJ_c}{M_c^2}. \quad (13.8)$$

Moreover,

$$\frac{V_n}{n} \implies \text{Uniform}[0, 1/M_c], \quad \frac{\mathbb{E}V_n}{n} \rightarrow \frac{1}{2M_c}, \quad \frac{\text{Var } V_n}{n^2} \rightarrow \frac{1}{12M_c^2}. \quad (13.9)$$

*Proof.* When  $u < u_c$ , the denominator in (13.1) does not vanish on  $|q| \leq \rho_2$ ; only the pole of  $B$  contributes. Uniformly for complex  $v$  near one,

$$\frac{f_n(uv)}{f_n(u)} \rightarrow \frac{1 - \bar{u}}{1 - \bar{u}v}.$$

This proves the geometric law. Marking the area outside the final block by an extra variable close to one leaves all exterior factors analytic at  $\rho_2$ ; the same residue ratio gives a finite exponential moment for that area, uniformly for large  $n$ .

For  $u > u_c$ , positivity and the consecutive kernel degrees give a unique simple pole  $r < \rho_2$ . Its coefficient is  $B(r)/M_1(r)$ . Analytically perturbing  $u$  to  $ue^s$  gives

$$\log \mathbb{E}_{n,u} e^{sV_n} = n \log \frac{r(u)}{r(ue^s)} + O(1),$$

with an analytic, uniformly controlled  $O(1)$  term. Differentiating  $\log K(r(ue^s)) = -\log u - s$  gives first and second derivatives  $1/M_1(r)$  and  $V_1(r)/M_1(r)^3$  for the leading logarithmic moment generating function. Substitution  $s = it/\sqrt{n}$  proves the central limit theorem. Since  $B$  is analytic beyond  $r$ , the final block is exponentially tight.

At criticality, put  $t = 1 - q/\rho_2$ . Then

$$1 - u_c K(q) = M_c t - J_c t^2 + O(t^3).$$

Using (13.3), the singular part of  $F$  is

$$\frac{b/M_c}{t^2} + \frac{b_0/M_c + bJ_c/M_c^2}{t}.$$

Coefficient extraction proves (13.7). The distributional limit is proved with its full scaling window in Theorem 14.1 below. Since every valley uses at least one part,  $0 \leq V_n/n \leq 1$ ; weak convergence therefore also implies convergence of the first two moments and the variance statement.  $\square$

[write] **Added 5 October 2026, batch 98.** The ordinary model  $u = 1$  lies in the supercritical regime, where  $r(1) = \rho$  and (13.6) is Theorem 6.1 with  $\mu = 1/M$ ,  $\sigma^2 = V/M^3$ ; the tightness of the final block at  $u = 1$  is part of Theorem 6.2. The regimes  $u \leq u_c$  and the supercritical regime at  $u \neq 1$  are new.

**Corollary 13.2** (A first-order transition). *The exponential growth rate is*

$$\Lambda(u) = \lim_{n \rightarrow \infty} \frac{1}{n} \log f_n(u) = \begin{cases} -\log \rho_2, & u \leq u_c, \\ -\log r(u), & u > u_c. \end{cases} \quad (13.10)$$

*Its derivative with respect to  $\log u$  jumps from 0 to  $1/M_c$  at  $u_c$ .*

*Proof.* The coefficient asymptotics give the formula. Implicit differentiation of  $uK(r(u)) = 1$  gives  $-d \log r / d \log u = 1/M_1(r)$ ; now let  $u \downarrow u_c$ .  $\square$

The critical value has a compact exact expression in the second unweighted pole:

$$u_c = \frac{\rho_2(1 - \rho_2)(1 - \rho_2^2)}{1 + \rho_2^4}. \quad (13.11)$$

Indeed,  $K_2(\rho_2) = 1$  gives  $U_3(\rho_2) = (1 - \rho_2^2)/\rho_2^2$ , and (12.2) then gives  $U_2(\rho_2) = (1 + \rho_2^4)/(\rho_2^2(1 - \rho_2^2))$ . Numerically,

$$\begin{aligned} \rho_2 &= 0.6285205937788026\dots, & u_c &= 0.1221810090081765\dots, \\ M_c &= 13.41194879035466\dots, & 1/M_c &= 0.07456038012307058\dots, \\ \kappa &= 0.2326297632869357\dots, & \kappa_0 &= -0.3327584620888633\dots \end{aligned}$$

The ordinary uniform model is  $u = 1 > u_c$ ; the new transition is a weighted refinement, not a claim that the ordinary model is critical.

## 14 The critical window and all finite-size orders

Define the entire function

$$G(z) = \int_0^1 e^{zx} dx = \begin{cases} (e^z - 1)/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (14.1)$$

The integral definition is preferable near  $z = 0$ .

**Theorem 14.1** (Uniform critical window). *Let  $u_n(s) = u_c e^{s/n}$ . Uniformly for  $s$  in any fixed compact subset of  $\mathbb{C}$ ,*

$$\rho_2^n f_n(u_n(s)) = \kappa n G(s/M_c) + G_1(s) + O(n^{-1}), \quad (14.2)$$

where

$$D_c = \frac{V_c}{2M_c^3}, \quad d_c = b \left( -\frac{1}{2} + \frac{1}{M_c} + \frac{J_c}{M_c^2} \right) + \frac{b_0}{M_c}, \quad (14.3)$$

$$G_1(s) = \frac{b}{2} + e^{s/M_c} (d_c + b D_c s). \quad (14.4)$$

For real  $s$ , under  $\mathbb{P}_{n, u_n(s)}$  the random variable  $V_n/n$  converges to the density

$$p_s(x) = \frac{M_c e^{sx}}{G(s/M_c)} \mathbf{1}_{[0, 1/M_c]}(x). \quad (14.5)$$

In particular  $p_0$  is the uniform density in (13.9).

*Proof.* For  $u$  near  $u_c$ , choose a fixed circle  $|q| = R > \rho_2$  inside which the only singularities of  $F(q, u)$  are the pole at  $\rho_2$  and the nearby root  $r(u)$  of  $uK(r) = 1$ . Their grouped contribution is analytic in  $u$ , including at coalescence. When the roots are distinct, residue extraction gives

$$\rho_2^n f_n(u_c e^\epsilon) = \frac{b}{1 - e^\epsilon} + \frac{B(r)}{M_1(r)} \exp\left(n \log \frac{\rho_2}{r}\right) + O((\rho_2/R)^n). \quad (14.6)$$

The error is uniform near  $\epsilon = 0$ : subtract the grouped principal parts, or equivalently use a fixed contour encircling both poles together.

Write  $r = \rho_2(1 - \delta)$ . Solving  $u_c K(\rho_2(1 - \delta)) = e^{-\epsilon}$  gives

$$\delta = \frac{\epsilon}{M_c} + \left(\frac{J_c}{M_c^3} - \frac{1}{2M_c}\right) \epsilon^2 + O(\epsilon^3). \quad (14.7)$$

Consequently

$$\log \frac{\rho_2}{r} = \frac{\epsilon}{M_c} + D_c \epsilon^2 + O(\epsilon^3), \quad \frac{B(r)}{M_1(r)} = \frac{b}{\epsilon} + d_c + O(\epsilon). \quad (14.8)$$

Here  $V_c = 2J_c + M_c - M_c^2$ . Also  $b/(1 - e^\epsilon) = -b/\epsilon + b/2 + O(\epsilon)$ . Insert  $\epsilon = s/n$  into (14.6); the singular terms cancel and give (14.2)–(14.4). The cancellation is holomorphic, so the expansion is uniform also at  $s = 0$  and at complex zeros of  $G$ . It is an additive estimate, not a relative estimate at such zeros.

For real  $s, t$ , the moment generating function of  $V_n/n$  is

$$\frac{f_n(u_c e^{(s+t)/n})}{f_n(u_c e^{s/n})} \rightarrow \frac{G((s+t)/M_c)}{G(s/M_c)}.$$

This is exactly the transform of (14.5). The bounded support of  $V_n/n$  makes transform convergence sufficient and gives moment convergence as well.  $\square$

**Proposition 14.2** (All-orders window expansion). *For every  $N \geq 1$  there are entire functions  $G_0, \dots, G_N$  such that, uniformly on fixed compact  $s$ -sets,*

$$\rho_2^n f_n(u_c e^{s/n}) = \sum_{j=0}^N n^{1-j} G_j(s) + O(n^{-N}). \quad (14.9)$$

Here  $G_0(s) = \kappa G(s/M_c)$ ,  $G_1$  is (14.4), and  $G_j(0) = 0$  for  $j \geq 2$ . Every coefficient is obtained by finite series arithmetic from the local Taylor coefficients of  $K$  and the Laurent coefficients of  $B$  at  $\rho_2$ .

*Proof.* Expand the analytic inverse root  $r(u_c e^\epsilon)$ , the Laurent series  $B(r)/M_1(r)$ , and  $b/(1 - e^\epsilon)$  in (14.6) to a prescribed finite order. The exponential  $\exp(n \log(\rho_2/r))$  is then expanded after  $\epsilon = s/n$ . At each order only finitely many Taylor coefficients are needed. The simple  $1/\epsilon$  terms cancel; all remaining coefficient functions are polynomials in  $s$  times  $e^{s/M_c}$ , together with a possible removable divided difference. They extend to entire functions. Analytic Taylor remainder bounds on a fixed  $\epsilon$ -disk prove the stated uniform remainder. At  $s = 0$ , (13.7) has exactly a degree-one polynomial prefactor plus an exponentially small error, forcing all later coefficients to vanish.  $\square$

This is a genuine uniform coalescing-pole expansion. Treating the two residues separately and ignoring their cancellation would give divergent prefactors and an incorrect critical approximation.

## 15 A hierarchy of multicritical Dirichlet laws

The one-weight transition is the first member of a finite-level hierarchy. Fix  $m \geq 1$ , mark valleys of heights  $1, \dots, m$ , and leave all higher valleys unweighted. Set

$$u_j^* = \frac{1}{K_j(\rho_{m+1})}, \quad M_j^* = M_j(\rho_{m+1}), \quad 1 \leq j \leq m. \quad (15.1)$$

Iteration of the weighted decomposition gives the exact identity

$$F_m(q, \mathbf{u}) = \sum_{\ell=1}^m H_\ell(q) \prod_{j=1}^{\ell-1} \frac{E_j(q)}{1 - u_j K_j(q)} + D_{m+1}(q) \prod_{j=1}^m \frac{E_j(q)}{1 - u_j K_j(q)}. \quad (15.2)$$

The last term is the branch that reaches level  $m+1$ . It has  $m+1$  critical factors; the earlier branches have at most  $m-1$ . [write: at  $m=1$ ,  $\rho_{m+1} = \rho_2$ ,  $u_1^* = u_c$ ,  $M_1^* = M_c$  and  $F_1 = F$  of (13.1).]

### 15.1 The elementary simplex mechanism

Write

$$\Delta_m = \{(\xi_1, \dots, \xi_m) : \xi_j \geq 0, \sum_{j=1}^m \xi_j \leq 1\}, \quad \xi_0 = 1 - \sum_{j=1}^m \xi_j.$$

Integration on  $\Delta_m$  means ordinary  $m$ -dimensional Lebesgue measure; its volume is  $1/m!$ .

**Lemma 15.1** (Nearby simple poles). *Suppose  $r_{i,n} = R_0 \exp(\eta_i/n + O(\log n/n^2))$ ,  $0 \leq i \leq m$ , where the real  $\eta_i$  range over a compact set. Let  $\Xi_n$  be analytic and uniformly bounded on a fixed disk  $|q| \leq R > R_0$ , with  $\Xi_n(R_0) = a_* + O(\log n/n)$ . Then*

$$\begin{aligned} [q^n] \left[ \Xi_n(q) \prod_{i=0}^m (1 - q/r_{i,n})^{-1} \right] \\ = R_0^{-n} n^m \left[ a_* \int_{\Delta_m} \exp \left( - \sum_{i=0}^m \eta_i \xi_i \right) d\boldsymbol{\xi} + O \left( \frac{\log n}{n} \right) \right]. \end{aligned} \quad (15.3)$$

If the root and amplitude errors omit  $\log n$ , so does the final error. Coincident  $\eta_i$  cause no exception.

*Proof.* First take  $\Xi_n = 1$ . The coefficient is the finite sum over  $k_0 + \dots + k_m = n$ ,  $k_i \geq 0$ , of  $\prod r_{i,n}^{-k_i}$ . Dividing by  $R_0^{-n}$  gives, uniformly for every summand,

$$\exp \left( - \sum_i \eta_i k_i / n \right) (1 + O(\log n/n)).$$

The grid sum divided by  $n^m$  is a Riemann sum on  $\Delta_m$ . Its integrand and first derivatives are uniformly bounded, so its error is  $O(1/n)$ . For general  $\Xi_n$ , convolve with its Taylor coefficients. Cauchy's bound makes their absolute values at radius  $R_0$  geometrically summable, including one extra factor of the index. Shifting  $n$  by that index changes the leading Riemann sum by  $O((k+1)n^{m-1})$ . Summation therefore multiplies the leading term by  $\Xi_n(R_0)$  and adds only  $O(R_0^{-n} n^{m-1})$ . The assumed error in  $\Xi_n(R_0)$  completes the proof.  $\square$

[write] **Added 5 October 2026, batch 98.** The manuscript states this lemma with its letters  $R$  for the common pole location and  $T$  for the radius, printed here as  $R_0$  and  $R$ ; it is applied at  $R_0 = \rho_{m+1}$ , and  $R_0 = \rho_2$  when  $m=1$ .

**Theorem 15.2** (Multicritical count and area sharing). *At the weights (15.1), for some  $R > \rho_{m+1}$ ,*

$$[q^n]F_m(q, \mathbf{u}^*) = \rho_{m+1}^{-n} \Pi_m(n) + O(R^{-n}), \quad (15.4)$$

where  $\Pi_m$  is a degree- $m$  polynomial with positive leading coefficient

$$\kappa^{(m)} = \frac{c_{m+1}}{m!} \prod_{j=1}^m \frac{E_j(\rho_{m+1})}{M_j^*}. \quad (15.5)$$

Let  $\mathcal{A}_{j,n}$  be the total area in repeated units at level  $j$ , let  $|T_n|$  be the area of the final  $\mathcal{D}_{m+1}$  block, and let  $V_{j,n}$  count height- $j$  valleys. On the dominant branch,

$$\left( \frac{\mathcal{A}_{1,n}}{n}, \dots, \frac{\mathcal{A}_{m,n}}{n}, \frac{|T_n|}{n} \right) \implies \text{Dirichlet}(1, \dots, 1), \quad (15.6)$$

with  $m+1$  coordinates, and

$$\frac{M_j^* V_{j,n} - \mathcal{A}_{j,n}}{n} \longrightarrow 0 \quad \text{in probability.} \quad (15.7)$$

The total endpoint-run area is tight. The probability of an earlier branch is  $O(n^{-2})$  (exponentially small when  $m = 1$ ).

*Proof.* At  $\rho_{m+1}$ , the tail  $D_{m+1}$  has one simple pole and each denominator in (15.2) has one simple zero. The highest Laurent coefficient is  $c_{m+1} \prod E_j(\rho_{m+1})/M_j^*$ , giving pole order  $m+1$  and (15.5). The other branches have pole order at most  $m-1$ . There are no other singularities on the first circle; subtraction of the finite Laurent principal part proves (15.4). This also proves the earlier-branch probability estimate.

Mark the area in the level- $j$  repeated units by replacing  $K_j(q)$  with  $K_j(qe^{v_j/n})$ , and mark terminal area by replacing  $D_{m+1}(q)$  with  $D_{m+1}(qe^{v_0/n})$ . Their poles move to  $\rho_{m+1}e^{-v_j/n}$  and  $\rho_{m+1}e^{-v_0/n}$ . The remaining analytic factors are uniformly bounded and converge to their original values. Applying Lemma 15.1 and normalizing gives the moment generating function

$$m! \int_{\Delta_m} \exp \left( \sum_{i=0}^m v_i \xi_i \right) d\xi,$$

which is that of the stated Dirichlet distribution.

If valley counts are simultaneously marked by  $e^{s_j/n}$ , the  $j$ th pole moves instead by logarithmic amount  $-(v_j + s_j/M_j^*)/n + O(n^{-2})$ . The joint limiting transform therefore identifies  $V_{j,n}/n$  with  $\xi_j/M_j^*$  and  $\mathcal{A}_{j,n}/n$  with  $\xi_j$ . This proves (15.7). Endpoint-run marking affects only analytic factors; a small fixed positive exponential mark is allowed because  $\rho_{m+1} < 1$ . The corresponding normalized coefficient ratio stays bounded, proving tightness with an exponential moment.  $\square$

[write] **Added 5 October 2026, batch 98.** At  $m = 1$ ,  $c_2 E_1(\rho_2) = c_2/(1 - \rho_2)^2 = b$  by Appendix C, so  $\kappa^{(1)} = b/M_c = \kappa$  and  $\Pi_1(n) = \kappa n + \kappa_0$ , consistent with (13.7); for  $m = 1$  there is a single earlier branch,  $H_1$ , which is analytic at  $\rho_2$ .

**Corollary 15.3** (Macroscopic covariance). *At the fully critical weights,*

$$\frac{\mathbb{E}V_{j,n}}{n} \longrightarrow \frac{1}{(m+1)M_j^*}, \quad (15.8)$$

$$\frac{\text{Var } V_{j,n}}{n^2} \longrightarrow \frac{m}{(m+1)^2(m+2)(M_j^*)^2}, \quad (15.9)$$

$$\frac{\text{Cov}(V_{i,n}, V_{j,n})}{n^2} \longrightarrow -\frac{1}{(m+1)^2(m+2)M_i^*M_j^*} \quad (i \neq j). \quad (15.10)$$

*Proof.* The count densities are bounded. Their moments converge to those of  $\xi_j/M_j^*$  under the uniform-simplex distribution. Integrating monomials on  $\Delta_m$  gives the displayed means and covariances.  $\square$

## 15.2 Finite-dimensional critical windows and phase chambers

For  $u_{j,n} = u_j^* e^{s_j/n}$ , the normalized leading coefficient is multiplied by

$$\mathcal{G}_m(\mathbf{s}) = m! \int_{\Delta_m} \exp\left(\sum_{j=1}^m \frac{s_j \xi_j}{M_j^*}\right) d\boldsymbol{\xi}. \quad (15.11)$$

The limiting area law is the Dirichlet law tilted by this exponential. The proof is the same nearby-pole argument. In particular the critical window has width  $1/n$  in every logarithmic weight coordinate.

**Proposition 15.4** ([write] The tilted multicritical window). *Let  $\mathbf{s}$  range over a compact subset of  $\mathbb{R}^m$  and  $u_{j,n} = u_j^* e^{s_j/n}$ . Then*

$$\frac{[q^n]F_m(q, \mathbf{u}_n)}{[q^n]F_m(q, \mathbf{u}^*)} = \mathcal{G}_m(\mathbf{s}) + O(n^{-1}),$$

and, under  $\mathbb{P}_{n, \mathbf{u}_n}$ , the vector in (15.6) converges in law to the probability measure with density proportional to  $\exp(\sum_{j=1}^m s_j \xi_j / M_j^*)$  on  $\Delta_m$  (with respect to  $d\boldsymbol{\xi}$ ), and (15.7) holds.

*Proof.* This proof was added in writing; the manuscript gives the sentence before the proposition. Each  $u_j^* K_j$  has a simple zero of  $1 - u_j^* K_j$  at  $\rho_{m+1}$ , with  $d \log(u_j^* K_j) / d \log q = M_j^* > 0$  there. By the analytic implicit function theorem, for  $\epsilon$  near 0 the equation  $u_j^* e^\epsilon K_j(r) = 1$  has a unique root near  $\rho_{m+1}$ ,  $\log(r/\rho_{m+1}) = -\epsilon/M_j^* + O(\epsilon^2)$ , and  $1 - u_j^* e^\epsilon K_j(q) = (1 - q/r)g_j(q, \epsilon)$  with  $g_j$  analytic and nonvanishing on a fixed disk  $|q| \leq R$ ,  $R > \rho_{m+1}$ , uniformly in  $\epsilon$  (choose  $R$  so that no other zero of  $1 - u_j^* K_j$  and no other pole of  $D_{m+1}$  lies in  $|q| \leq R$ ; both sets are locally finite, and  $\rho_{m+1}$  is the only singular point on  $|q| = \rho_{m+1}$  by the triangle-inequality argument of Lemma 12.2). Write  $D_{m+1}(q) = (1 - q/\rho_{m+1})^{-1} g_0(q)$  likewise. With  $\epsilon = s_j/n$ , the dominant branch of (15.2) is  $\Xi_n(q) \prod_{i=0}^m (1 - q/r_{i,n})^{-1}$  with  $r_{0,n} = \rho_{m+1}$ ,  $\eta_0 = 0$ ,  $\eta_j = -s_j/M_j^*$ , root errors  $O(n^{-2})$  and  $\Xi_n = g_0 \prod_j E_j/g_j$ , whose value at  $\rho_{m+1}$  is  $a_* + O(1/n)$  with  $a_* = c_{m+1} \prod_j E_j(\rho_{m+1})/M_j^* = m! \kappa^{(m)}$ . Lemma 15.1 (without logarithms) gives  $\rho_{m+1}^{-n} n^m [a_* \int_{\Delta_m} \exp(\sum_j s_j \xi_j / M_j^*) d\boldsymbol{\xi} + O(1/n)]$ . The earlier branches have at most  $m-1$  such factors, and the same lemma bounds their coefficients by  $O(\rho_{m+1}^{-n} n^{m-2})$ . Dividing by the case  $\mathbf{s} = 0$ , whose integral is  $1/m!$ , gives the first claim. Adding the area marks  $v_j/n$ ,  $v_0/n$  of the proof of Theorem 15.2 shifts  $\eta_i$  by  $-v_i$ ; the ratio of the marked to the unmarked coefficient converges to  $\int \exp(\sum_i v_i \xi_i + \sum_j s_j \xi_j / M_j^*) d\boldsymbol{\xi} / \int \exp(\sum_j s_j \xi_j / M_j^*) d\boldsymbol{\xi}$  for all real  $\mathbf{v}$ , which is the moment generating function of the tilted law; the variables are bounded, so this identifies the limit. The count-area identification is that of Theorem 15.2, with  $\eta_j$  shifted by the constant  $-s_j/M_j^*$ .  $\square$

There is also a complete fixed-parameter classification. For arbitrary positive weights  $u_1, \dots, u_m$ , let  $r_j$  solve  $u_j K_j(r_j) = 1$  and set

$$r_* = \min(r_1, \dots, r_m, \rho_{m+1}). \quad (15.12)$$

If precisely  $\nu$  of these  $m+1$  values equal  $r_*$ , then the dominant pole has order  $\nu$  and the coefficients have form  $r_*^{-n}$  times a degree- $(\nu-1)$  polynomial, plus an exponentially smaller

error. To see noncancellation, take the suffix after the last active weighted level: all its inactive denominators are positive and finite at  $r_*$ . Its positive value, or its positive tail residue, is multiplied by exactly the active simple-pole factors. Earlier branches cannot remove this highest-order positive contribution. Thus the phase boundaries are the explicit equal-root hypersurfaces, not hypothetical singularities inferred from a numerical phase plot.

**Proposition 15.5** ([write] Phase chambers). *In the setting of (15.12),  $r_*$  is the only singularity of  $F_m(q, \mathbf{u})$  on  $|q| \leq r_*$ ; it is a pole of order exactly  $\nu$ , and for some  $R > r_*$ ,  $[q^n]F_m(q, \mathbf{u}) = r_*^{-n}\Pi(n) + O(R^{-n})$  with a polynomial  $\Pi$  of degree  $\nu - 1$  and positive leading coefficient.*

*Proof.* This proof was added in writing, expanding the manuscript's sketch above. Each  $r_j \in (0, 1)$  exists and is unique, because  $K_j$  increases from 0 to  $\infty$  on  $(0, 1)$ . Call an index active if its value in (15.12) equals  $r_*$  (the index  $m + 1$  standing for  $\rho_{m+1}$ ). For  $|q| \leq r_*$  and  $j \leq m$ ,  $|u_j K_j(q)| \leq u_j K_j(|q|) \leq u_j K_j(r_j) = 1$ , with equality only if  $|q| = r_j = r_*$ , and then, by the consecutive degrees  $2j + 1, 2j + 2$  in the support of  $K_j$  (Lemma 12.2), only at  $q = r_*$ . So  $1 - u_j K_j$  has no zero on  $|q| \leq r_*$  for inactive  $j$  and only the simple zero  $r_*$  for active  $j$  (simple because  $K'_j(r_*) > 0$ ). By Lemma 12.2,  $D_{m+1}$  is analytic on  $|q| \leq r_*$  unless  $m + 1$  is active, and then its only singularity there is the simple pole  $\rho_{m+1} = r_*$  with residue amplitude  $c_{m+1} > 0$ . The functions  $H_\ell, E_j$  are analytic on the unit disk. This proves the first claim, and the order of the pole of each branch of (15.2) is the number of active indices among its factors:  $\#(\mathcal{I} \cap \{1, \dots, \ell - 1\})$  for the branch  $\ell \leq m$ , and  $\nu$  for the final branch, where  $\mathcal{I}$  is the active set. So the order of  $F_m$  at  $r_*$  is at most  $\nu$ , and the coefficient of  $(1 - q/r_*)^{-\nu}$  is the sum, over the branches attaining order  $\nu$ , of products of:  $H_\ell(r_*) > 0$  or the value  $D_{m+1}(r_*) > 0$  (inactive  $m + 1$ ) or the amplitude  $c_{m+1} > 0$  (active  $m + 1$ );  $E_j(r_*) > 0$ ;  $1/(1 - u_j K_j(r_*)) > 0$  for inactive  $j$  (since  $u_j K_j(r_*) < u_j K_j(r_j) = 1$ ); and  $1/M_j(r_*) > 0$  for active  $j$ , the leading coefficient of  $1/(1 - u_j K_j(q))$  at  $r_*$  because  $u_j r_* K'_j(r_*) = M_j(r_*)$  when  $u_j K_j(r_*) = 1$ . The final branch always attains order  $\nu$ , so this sum is positive, the order is exactly  $\nu$ , and no cancellation occurs. Poles of the factors are locally finite, so a radius  $R > r_*$  with no other singularity on  $|q| \leq R$  exists; subtracting the principal part at  $r_*$  and applying Cauchy's bound on  $|q| = R$  gives the expansion, with  $[q^n](1 - q/r_*)^{-k} = \binom{n+k-1}{k-1} r_*^{-n}$  giving the polynomial of degree  $\nu - 1$ .  $\square$

[write: at  $m = 1$  the chambers are those of Theorem 13.1:  $\nu = 1$  off  $u_1 = u_c$  and  $\nu = 2$  at  $u_1 = u_c$ , where  $r_1 = \rho_2$ .]

## 16 An exact height-tail identity and its exponential expansion

Let  $U_j^{\leq h}$  and  $D_j^{\leq h}$  count the same objects with every part at most  $h$ . For  $h \geq j$ ,

$$U_j - U_j^{\leq h} = U_{h+1} \prod_{r=j}^h (1 - q^r)^{-2}. \quad (16.1)$$

This is simply the tail of the maximum-part decomposition. Put

$$P_j(q) = \prod_{r=j}^{\infty} (1 - q^r)^{-2}, \quad P_1(q) = \prod_{r=1}^{\infty} (1 - q^r)^{-2}. \quad (16.2)$$

These products are analytic and nonzero for  $|q| < 1$ . [write: they are the products  $P_j$  of Section 7, there used for  $0 < q < 1$ ; they are not the pole sets  $\mathcal{P}_R$  of Theorem 5.1.]

**Lemma 16.1** (An analytic two-variable tail). *For  $|q| < 1$  define*

$$\Psi(q, z) = \sum_{k \geq 0} zq^k (1 - zq^k) \prod_{r=k+1}^{\infty} (1 - zq^r)^2. \quad (16.3)$$

*The series is analytic in  $(q, z)$  locally for  $|q| < 1$ , including near  $z = 0$ , and*

$$U_j(q) - U_j^{\leq h}(q) = P_j(q) \Psi(q, q^{h+1}). \quad (16.4)$$

*Its first coefficients are*

$$\Psi(q, z) = \frac{z}{1-q} - \frac{z^2}{(1-q)^2} + O(z^3). \quad (16.5)$$

*All coefficients are explicitly computable rational functions of  $q$ .*

*Proof.* On compact  $|q| \leq r < 1$ ,  $|z| \leq Z$ , the infinite products are uniformly bounded, while the summands are  $O(r^k)$ . This proves normal convergence. Multiply the maximum-part sum for  $U_{h+1}$  by  $\prod_{r=h+1}^{\infty} (1 - q^r)^2$  and set  $z = q^{h+1}$ ; the resulting summands are exactly those in (16.3). Equation (16.1) then proves (16.4).

For an explicit coefficient procedure, write

$$\prod_{r=1}^{\infty} (1 - zq^r)^2 = \sum_{\ell \geq 0} \gamma_{\ell}(q) z^{\ell}, \quad \gamma_{-1} = 0.$$

Summing the geometric series in  $q^{k\ell}$  in (16.3) gives

$$[z^{\ell}] \Psi(q, z) = \frac{\gamma_{\ell-1}(q) - \gamma_{\ell-2}(q)}{1 - q^{\ell}} \quad (\ell \geq 1), \quad (16.6)$$

with  $\gamma_{-2} = 0$  if needed. More explicitly,

$$\gamma_{\ell}(q) = (-1)^{\ell} \sum_{a+b=\ell} \frac{q^{a(a+1)/2+b(b+1)/2}}{(q; q)_a (q; q)_b}, \quad (q; q)_a = \prod_{r=1}^a (1 - q^r). \quad (16.7)$$

For completeness, the coefficient formula for one product follows by writing  $Q(z) = \prod_{r \geq 1} (1 - zq^r) = (1 - zq)Q(qz)$ : comparison of coefficients recursively gives  $[z^a]Q = (-1)^a q^{a(a+1)/2} / (q; q)_a$ . Squaring proves (16.7). Since  $\gamma_0 = 1$  and  $\gamma_1 = -2q/(1 - q)$ , (16.5) follows.  $\square$

[write: in this proof  $Z$  is a bound and  $Q(z)$  a local product; neither is Part I's  $Z$  or  $Q_j$ , and the summation indices  $a, b$  are local.]

Set  $K_j^{\leq h} = H_j U_{j+1}^{\leq h}$ . The lemma implies, locally uniformly for real  $q$  in a compact subinterval of  $(0, 1)$ ,

$$K_j(q) - K_j^{\leq h}(q) = a_j(q) q^h + O(q^{2h}), \quad a_j(q) = \frac{q^{j+1} P_{j+1}(q)}{(1 - q^j)(1 - q)}. \quad (16.8)$$

Each differentiation adds at most a factor of order  $h$  to the error scale. In particular

$$K(q) - K^{\leq h}(q) = P_1(q) q^{h+2} - \frac{P_1(q) q^{2h+3}}{1 - q} + O(q^{3h}). \quad (16.9)$$

## 16.1 The height-cutoff pole to all exponential orders

Write

$$\rho = \rho_1, \quad M_\rho = M_1(\rho), \quad V_\rho = V_1(\rho), \quad \alpha = \frac{\rho^2 P_1(\rho)}{M_\rho}, \quad L = \log(1/\rho). \quad (16.10)$$

[write: so  $M_\rho = M$  and  $V_\rho = V$  of (6.3), and  $L = \log \lambda$  is Part I's  $L$  of (8.1).] For sufficiently large integer  $h$ , let  $\hat{\rho}_h$  be the root of  $K^{\leq h}(\hat{\rho}_h) = 1$  near  $\rho$ ; it is the dominant pole of  $D_1^{\leq h}$ . The hat distinguishes a maximum-height cutoff from the minimum-part poles  $\rho_j$ . Define

$$\delta_h = \log(\hat{\rho}_h/\rho).$$

**Theorem 16.2** (Height-cutoff exponential expansion). *There are polynomials  $\varpi_{r-1}(h)$  of degree at most  $r-1$  such that, for every fixed  $N \geq 1$ ,*

$$\delta_h = \sum_{r=1}^N \varpi_{r-1}(h) \rho^{rh} + O_N(h^N \rho^{(N+1)h}). \quad (16.11)$$

The first two terms are

$$\delta_h = \alpha \rho^h + \left[ \alpha^2(h+g) - \frac{\alpha \rho}{1-\rho} \right] \rho^{2h} + O(h^2 \rho^{3h}), \quad (16.12)$$

$$g = 2 + 2 \sum_{k \geq 1} \frac{k \rho^k}{1-\rho^k} - \frac{M_\rho^2 + V_\rho}{2M_\rho}. \quad (16.13)$$

The leading coefficient of every polynomial is

$$[h^{r-1}] \varpi_{r-1}(h) = \frac{r^{r-1}}{r!} \alpha^r. \quad (16.14)$$

Thus the top-degree terms are precisely the expansion of  $-W_0(-\alpha h \rho^h)/h$ .

*Proof.* Let  $q = \rho e^\delta$  and  $\epsilon = \rho^h$ . The root equation is

$$K(\rho e^\delta) - 1 = \sum_{r \geq 1} A_r(\rho e^\delta) \epsilon^r e^{rh\delta},$$

where the analytic  $A_r$  are obtained from (16.6) (Appendix D); in particular  $A_1(q) = q^2 P_1(q)$  and  $A_2(q) = -q^3 P_1(q)/(1-q)$ . At  $\delta = 0$  the derivative of the left side is  $M_\rho > 0$ . For small  $h\epsilon$ , the implicit equation has a unique solution  $\delta = O(\epsilon)$ , and its derivative remains bounded away from zero.

Substitute a formal series in  $\epsilon$  and determine its coefficients successively. At order  $r$ , the unknown coefficient appears only as  $M_\rho \varpi_{r-1}$ . Taylor differentiation of  $e^{rh\delta}$  shows inductively that its remaining right side is a polynomial in  $h$  of degree at most  $r-1$ . The Taylor residual after  $N$  terms is  $O_N(h^N \epsilon^{N+1})$ : all derivatives of the analytic  $A_r$  and of  $K$  are bounded on a fixed neighborhood, and the exponential factors are bounded because  $h\delta \rightarrow 0$ . Division by the nonvanishing implicit derivative converts this residual into the claimed root error.

For the first two coefficients, use

$$K(\rho e^\delta) - 1 = M_\rho \delta + \frac{M_\rho^2 + V_\rho}{2} \delta^2 + O(\delta^3),$$

and  $\rho A'_1(\rho)/A_1(\rho) = 2 + 2 \sum k \rho^k / (1 - \rho^k)$ . This gives (16.12). To obtain degree  $r-1$  in  $h$ , retain only the linear term on the left and  $A_1(\rho) \epsilon e^{h\delta}$  on the right. Any nonlinear analytic term, derivative of  $A_1$ , or  $A_r$  with  $r \geq 2$  lowers the possible degree. The reduced equation  $\delta = \alpha \epsilon e^{h\delta}$  has solution  $-W_0(-\alpha h \epsilon)/h$ , whose Lagrange expansion yields (16.14).  $\square$

The theorem is an all-orders statement in the ordered scales  $\rho^{rh}$  multiplied by polynomials in  $h$ . It does not assert convergence of an unrestricted infinite asymptotic series, nor any Borel or Stokes property. The small argument of  $W_0$  is on its analytic branch at zero.

[write] **Added 5 October 2026, batch 98.** The manuscript writes  $a_r(q)$  for the order- $r$  coefficients in this proof and  $A_r(q)$  for the same functions in its Appendix B; they are printed as  $A_r$  in both places, since  $a_j(q)$  of (16.8) is a different family ( $a_1 = A_1$ , but  $a_2 \neq A_2$ ). The manuscript's ledger calls this expansion a “cutoff-pole transseries”; it is a finite-order expansion in the scales  $h^k \rho^{rh}$  with no convergence, Borel or Stokes claim (as the paragraph above says), and is not called a transseries here, as for Theorem 7.3 in Part I.

## 17 Tallest columns: a lattice-Gumbel law

Let  $H_n$  be the maximum part of a uniformly chosen unweighted valley-monotone composition of area  $n$ .

**Theorem 17.1** (Uniform maximum-height law). *Uniformly for integers  $h = h(n)$  satisfying  $0 < \tau_- \leq n\rho^h \leq \tau_+ < \infty$ , with fixed  $\tau_\pm$ ,*

$$\mathbb{P}(H_n \leq h) = \exp(-\alpha n\rho^h) \left( 1 + O_{\tau_\pm} \left( \frac{\log n}{n} \right) \right), \quad (17.1)$$

where  $\alpha$  is defined in (16.10). Numerically,

$$\begin{aligned} \rho &= 0.5059651577008067\dots, & \alpha &= 0.4502021724631094\dots, \\ L^{-1} &= 1.4678091752740682\dots \end{aligned} \quad (17.2)$$

*Proof.* Choose a fixed radius  $R$  strictly between  $\rho$  and  $\rho_2$ , close enough to  $\rho$  that  $1 - K$  has only its simple root there. Uniform convergence of  $U_2^{\leq h}$  to  $U_2$  on this disk and Rouché's theorem show that  $1 - K^{\leq h}$  has exactly one nearby root, the cutoff root of Theorem 16.2. The higher-level truncated functions  $D_2^{\leq h}$  are uniformly analytic on this disk, by the same normally convergent decomposition used in Lemma 12.2. Consequently

$$[q^n]D_1^{\leq h} = \hat{c}_h \hat{\rho}_h^{-n} + O(R^{-n}) \quad (17.3)$$

with a uniform error for all sufficiently large  $h$ .

We record a quantitative estimate for the numerator, rather than assume it. At a fixed real  $q < \rho_2$ , subtraction of the two level recurrences gives an error transported by the multipliers  $E_j/(1 - K_j)$ , whose products are uniformly bounded for  $j \geq 2$ . The additive kernel perturbation at level  $j$  is bounded by a constant times

$$D_{j+1}(q) H_j(q) P_{j+1}(q) q^{h+1} = O(q^{2j+h}).$$

This is summable in  $j$ , while the final omitted tail has size  $O(q^h)$ . It follows that  $D_2 - D_2^{\leq h} = O(q^h)$ ; differentiating the recurrences gives  $O(hq^h)$  for the first derivative. The same estimates hold locally with absolute values. Together with the kernel derivative estimates, they imply

$$\hat{c}_h/c_1 = 1 + O(h\rho^h). \quad (17.4)$$

Evaluation at the displaced root does not change this order because  $\hat{\rho}_h - \rho = O(\rho^h)$  and  $h\rho^h \rightarrow 0$ .

Divide (17.3) by (12.6) at  $j = 1$ . Theorem 16.2 gives  $n \log(\hat{\rho}_h/\rho) = \alpha n\rho^h + O(nh\rho^{2h})$ . In the specified window,  $h = O(\log n)$  and  $\rho^h \asymp n^{-1}$ . Both the exponential error and the residue-ratio error are therefore  $O(\log n/n)$ ; the uniform Cauchy error is exponentially smaller.  $\square$

[write] **Added 5 October 2026, batch 98.** The residue-ratio estimate (17.4) is argued in outline (“differentiating the recurrences gives  $O(hq^h)$ ”); the write records it as complete in outline, not as an open question. The theorem answers the second half of Part I’s question “Joint limits and extreme heights” (Section 10).

## 17.1 The lattice phase must not be discarded

Define

$$\beta_n = \frac{\log(\alpha n)}{L}. \quad (17.5)$$

If a subsequence satisfies  $\{\beta_n\} \rightarrow \theta \in [0, 1]$ , then, at every integer  $k$ ,

$$\mathbb{P}(H_n - \lfloor \beta_n \rfloor \leq k) \rightarrow \exp(-\rho^{k-\theta}). \quad (17.6)$$

These formulas define lattice distributions; when  $\theta = 1$ , the result is understood as the one-sided endpoint phase. Different phases generally give different laws. The centered maximum therefore does not have a single phase-independent continuous Gumbel limit. Tightness follows by taking fixed large positive and negative  $k$  in (17.6); in particular  $H_n = (\log n)/L + O_{\mathbb{P}}(1)$ .

**Theorem 17.2** (Gaussian counts and extremes are independent). *Put  $\mu = 1/M_\rho$  and  $\sigma^2 = V_\rho/M_\rho^3$ . Along any phase subsequence as above,*

$$\left( \frac{V_n - \mu n}{\sigma \sqrt{n}}, H_n - \lfloor \beta_n \rfloor \right) \Rightarrow (\chi, Y_\theta), \quad (17.7)$$

where  $\chi$  is standard normal,  $Y_\theta$  has (17.6), and the two limits are independent.

*Proof.* Mark the valley count by  $u = e^{s/\sqrt{n}}$  and impose the height cutoff. The untruncated root  $r(u)$  is analytic near  $u = 1$  and differs from  $\rho$  by  $O(n^{-1/2})$ . The truncated-root displacement is, uniformly in this neighborhood,

$$\log \frac{r_h(u)}{r(u)} = \frac{u a_1(r(u))}{M_1(r(u))} r(u)^h + O(hr(u)^{2h}).$$

For  $h = \beta_n + O(1)$ ,  $r(u)^h/\rho^h = 1 + O(\log n/\sqrt{n})$  and the prefactor converges to  $\alpha$ . The marked coefficient ratio is therefore the ordinary quasi-power transform for  $V_n$ , multiplied by  $\exp(-\alpha n \rho^h)(1 + o(1))$ . Set  $s = it/\sigma$  and include the centering factor. The first term tends to  $e^{-t^2/2}$  and the second tends to the lattice distribution function. Tightness of both coordinates and these mixed transforms identify the product limit.  $\square$

[write:  $\mu, \sigma^2$  are those of Theorem 6.1, and  $r(u)$  is the root used in its proof.]

## 18 Critical extreme heights are Dirichlet mixtures

The ordinary Gumbel law does not remain unchanged at a critical point. Several macroscopic regions coexist, and each has its own rate of producing a column above the cutoff.

Fix  $m$  as in Section 15. Define

$$\alpha_j^* = \frac{u_j^* a_j(\rho_{m+1})}{M_j^*} \quad (1 \leq j \leq m), \quad \alpha_0^* = \frac{a_{m+1}(\rho_{m+1})}{M_{m+1}(\rho_{m+1})}, \quad (18.1)$$

where  $a_j$  is the explicit product in (16.8). Index zero refers to the final  $\mathcal{D}_{m+1}$  block, whose unmarked dominant kernel is  $K_{m+1}$ .

**Theorem 18.1** (Joint critical window and height cutoff). *Let  $\mathbf{s}$  range over a fixed compact subset of  $\mathbb{R}^m$ , set  $u_{j,n} = u_j^* e^{s_j/n}$ , and let integers  $h = h(n)$  satisfy  $0 < \tau_- \leq \tau_n := n\rho_{m+1}^h \leq \tau_+ < \infty$ . Then*

$$\begin{aligned} \mathbb{P}_{n,u_n}(H_n \leq h) &= \frac{\int_{\Delta_m} \exp\left(\sum_{j=1}^m \frac{s_j \xi_j}{M_j^*} - \tau_n \sum_{i=0}^m \alpha_i^* \xi_i\right) d\xi}{\int_{\Delta_m} \exp\left(\sum_{j=1}^m \frac{s_j \xi_j}{M_j^*}\right) d\xi} + O\left(\frac{\log n}{n}\right). \end{aligned} \quad (18.2)$$

At exact criticality this is the Laplace transform of  $\sum_{i=0}^m \alpha_i^* X_i$ , where  $(X_0, \dots, X_m) \sim \text{Dirichlet}(1, \dots, 1)$ , evaluated at  $\tau_n$ .

*Proof.* Truncate every unimodal and valley-monotone factor at height  $h$  in (15.2). For a fixed marked level  $j \leq m$ , combine (16.8) with the implicit derivative at  $\rho_{m+1}$ . Its nearby root satisfies

$$\log \frac{r_{j,n,h}}{\rho_{m+1}} = \alpha_j^* \rho_{m+1}^h - \frac{s_j}{M_j^* n} + O\left(h\rho_{m+1}^{2h} + \frac{h\rho_{m+1}^h}{n} + n^{-2}\right). \quad (18.3)$$

Indeed, the derivative of  $\log(u_j^* K_j(q))$  with respect to  $\log q$  is  $M_j^*$ ; the kernel loss is  $u_j^* a_j(\rho_{m+1}) \rho_{m+1}^h$  to first order, and the logarithmic weight increment is  $s_j/n$ . Taylor's theorem and the derivative bound following (16.8) give the error.

The truncated terminal function  $D_{m+1}^{\leq h}$  has a nearby simple pole whose logarithmic displacement is  $\alpha_0^* \rho_{m+1}^h + O(h\rho_{m+1}^{2h})$ . Higher-level denominators are uniformly analytic on a fixed disk slightly larger than  $\rho_{m+1}$ . After factoring out these  $m+1$  nearby simple poles, the analytic numerator is uniformly bounded on that disk. At  $\rho_{m+1}$  it differs from its critical untruncated value by  $O(h\rho_{m+1}^h + n^{-1})$ , by the recurrence estimate in the proof of Theorem 17.1. This estimate also handles the possible collision of some of the moved poles: the analytic factorization does not divide by their mutual differences.

In the stated window,  $\rho_{m+1}^h \asymp n^{-1}$  and  $h = O(\log n)$ , so (18.3) has the form required by Lemma 15.1, with

$$\eta_j = \tau_n \alpha_j^* - s_j/M_j^*, \quad \eta_0 = \tau_n \alpha_0^*.$$

Earlier branches in (15.2) have at most  $m-1$  nearby pole factors and contribute  $O(n^{-2})$  relatively (or an exponentially small amount for  $m=1$ ). Apply the lemma to the dominant branch and divide by the corresponding untruncated coefficient with  $\eta_0 = 0$  and  $\eta_j = -s_j/M_j^*$ . The denominator integral is bounded away from zero on every compact real  $\mathbf{s}$ -set. This proves (18.2).  $\square$

[write] **Added 5 October 2026, batch 98.** The manuscript abbreviates  $R = R_m$  in this section; every such  $R$  is printed as  $\rho_{m+1}$ , and its window bounds  $a, b$  as  $\tau_{\pm}$ . The intensities  $\alpha_j^*$  are not the Lambert coefficients  $\alpha_k(c)$  of (7.11).

## 18.1 The first critical point in closed form

When  $m=1$ , let  $\xi_1$  be the macroscopic fraction of area in the bottom-level renewal blocks. Then  $\xi_1$  is uniform on  $[0, 1]$  at criticality. Writing  $\alpha_1^*$  and  $\alpha_0^*$  as in (18.1), the critical-window height law becomes

$$\mathbb{P}_{n, u_c e^{s/n}}(H_n \leq h) = \frac{e^{-\tau_n \alpha_0^*} G(s/M_c - \tau_n(\alpha_1^* - \alpha_0^*))}{G(s/M_c)} + O(\log n/n). \quad (18.4)$$

At  $s = 0$ , this simplifies to

$$\frac{e^{-\tau_n \alpha_1^*} - e^{-\tau_n \alpha_0^*}}{\tau_n(\alpha_0^* - \alpha_1^*)} + O(\log n/n), \quad (18.5)$$

with the continuous extension when the two intensities coincide. The numerical constants are

$$\begin{aligned} \alpha_1^* &= 0.3053024072567406 \dots, \\ \alpha_0^* &= 0.3765567576528582 \dots, \\ \frac{1}{\log(1/\rho_2)} &= 2.1533787748784431 \dots \end{aligned}$$

Thus critical maximum height has scale  $2.1533 \dots \log n$ , compared with  $1.4678 \dots \log n$  in the ordinary unweighted model.

The limiting expression is not a single exponential  $e^{-c\tau}$  unless all active intensities coincide. Indeed, the second Taylor coefficient would otherwise force  $\mathbb{E}(\sum \alpha_i^* X_i)^2 = (\mathbb{E} \sum \alpha_i^* X_i)^2$ , which fails for a nonconstant linear function on the simplex. This distinguishes an actual mixture from merely replacing the Gumbel intensity by its mean.

For the lattice statement, take  $\beta_n^{\text{crit}} = \log n / \log(1/\rho_{m+1})$  and a subsequence with fractional parts tending to  $\theta$ . At integer  $k$ , put  $\tau = \rho_{m+1}^{k-\theta}$  in (18.2). The resulting distribution function is the phase-dependent Dirichlet mixture. Joint marking of the area fractions shows more: conditional on their limiting values  $X_i$ , the extreme-height distribution has intensity  $\sum \alpha_i^* X_i$ . Consequently the maximum is generally *not* asymptotically independent of the critical macroscopic fractions, unlike the Gaussian independence in Theorem 17.2.

**Proposition 18.2** ([write] Conditional intensity of the critical maximum). *In the setting of Theorem 18.1, let  $\mathbf{X}_n$  be the vector in (15.6) on the dominant branch (and  $\mathbf{X}_n = 0$  on the earlier branches), and let  $\pi_s$  be the probability measure on the simplex with density proportional to  $\exp(\sum_{j=1}^m s_j \xi_j / M_j^*)$ . Along any subsequence with  $\tau_n \rightarrow \tau \in [\tau_-, \tau_+]$ , for every continuous function  $f$  on  $[0, 1]^{m+1}$ ,*

$$\mathbb{E}_{n, u_n}[f(\mathbf{X}_n); H_n \leq h] \longrightarrow \int f(\boldsymbol{\xi}) \exp\left(-\tau \sum_{i=0}^m \alpha_i^* \xi_i\right) \pi_s(d\boldsymbol{\xi}). \quad (18.6)$$

Thus, in the limit,  $\mathbf{X}_n$  has law  $\pi_s$  and, conditionally on  $\mathbf{X} = \boldsymbol{\xi}$ , the event  $\{H_n \leq h\}$  has probability  $\exp(-\tau \sum_i \alpha_i^* \xi_i)$ . The limits of  $\mathbf{X}_n$  and of  $\mathbf{1}\{H_n \leq h\}$  are independent if and only if  $\alpha_0^* = \alpha_1^* = \dots = \alpha_m^*$ .

*Proof.* This proof was added in writing; the manuscript asserts the statement in the paragraph above. Mark areas as in the proof of Theorem 15.2, by  $v_j/n$  in the level- $j$  repeated units and  $v_0/n$  in the final block, in the truncated decomposition of the proof of Theorem 18.1. Replacing  $q$  by  $qe^{v_j/n}$  in  $u_{j,n}K_j^{\leq h}$  multiplies its root  $r_{j,n,h}$  by exactly  $e^{-v_j/n}$ , and likewise for the cutoff pole of  $D_{m+1}^{\leq h}$ ; the remaining analytic factors, evaluated at  $qe^{v/n}$ , are uniformly bounded and change by  $O(1/n)$  at  $\rho_{m+1}$ . Lemma 15.1 applies with  $\eta_j = \tau_n \alpha_j^* - s_j/M_j^* - v_j$  and  $\eta_0 = \tau_n \alpha_0^* - v_0$ , and the earlier branches contribute  $O(n^{-2})$  relatively, as in that proof. Dividing by the unmarked untruncated coefficient gives, uniformly for  $\mathbf{v}$  in compact subsets of  $\mathbb{R}^{m+1}$ ,

$$\mathbb{E}_{n, u_n}[e^{\langle \mathbf{v}, \mathbf{X}_n \rangle}; H_n \leq h] = \int e^{\langle \mathbf{v}, \boldsymbol{\xi} \rangle} e^{-\tau_n \langle \boldsymbol{\alpha}^*, \boldsymbol{\xi} \rangle} \pi_s(d\boldsymbol{\xi}) + O\left(\frac{\log n}{n}\right),$$

with  $\langle \mathbf{v}, \boldsymbol{\xi} \rangle = \sum_{i=0}^m v_i \xi_i$ . The left side is the Laplace transform of the finite measure  $\mu_n(B) = \mathbb{P}_{n, u_n}(\mathbf{X}_n \in B, H_n \leq h)$  on the compact set  $[0, 1]^{m+1}$ . As  $\tau_n \rightarrow \tau$ , these transforms converge for every real  $\mathbf{v}$  to that of  $\mu(d\boldsymbol{\xi}) = e^{-\tau \langle \boldsymbol{\alpha}^*, \boldsymbol{\xi} \rangle} \pi_s(d\boldsymbol{\xi})$ , whose total mass is positive. Hence the total masses converge, and the normalized probability measures, whose moment generating

functions converge everywhere, converge weakly; so  $\mu_n \rightarrow \mu$  weakly, which is (18.6). With Proposition 15.4 ( $\mathbf{X}_n \Rightarrow \pi_s$ ) this identifies the joint limit as stated. It is a product law exactly when  $e^{-\tau\langle \alpha^*, \xi \rangle}$  is  $\pi_s$ -almost surely constant. Since  $\pi_s$  has a positive continuous density on the simplex and  $\langle \alpha^*, \xi \rangle = \alpha_0^* + \sum_{j=1}^m (\alpha_j^* - \alpha_0^*) \xi_j$  there, with  $\tau > 0$ , this happens if and only if all  $\alpha_i^*$  are equal.  $\square$

[write: at the first critical point the numerical values  $\alpha_1^* = 0.3053\dots$  and  $\alpha_0^* = 0.3766\dots$  above differ, so the critical maximum is not asymptotically independent of the area fraction there; these decimals are diagnostics, not interval certificates.]

## 19 Inversion of the new asymptotic expansions

### 19.1 Inverting the height-cutoff pole

A discrete cutoff has no automatic real inverse. Here an exact analytic interpolation is available: for large real  $h$  and real  $q$  near  $\rho$ , define

$$\widehat{K}_h(q) = K(q) - H_1(q)P_2(q)\Psi(q, q^{h+1}). \quad (19.1)$$

It agrees with the height-truncated kernel at every sufficiently large integer  $h$ . Its nearby root defines  $\delta(h)$  as in Theorem 16.2. The derivative of the leading tail with respect to  $h$  has the sign of  $\log \rho < 0$ , while subsequent derivatives are smaller, so  $\delta(h)$  is strictly decreasing for large  $h$ .

**Lemma 19.1** ([write] The interpolated shift decreases). *For all sufficiently large real  $h$ ,  $\widehat{K}_h(q) = 1$  has a unique root  $q = \rho e^{\delta(h)}$  with  $\delta(h)$  small;  $\delta$  is real analytic,  $\delta(h) = \alpha \rho^h (1 + O(h\rho^h))$ , and  $\delta'(h) < 0$ . At integers  $h \geq 2$ ,  $\widehat{K}_h = K^{\leq h}$  and  $\delta(h) = \delta_h$ .*

*Proof.* This proof was added in writing, expanding the one-line justification above. By (16.4) with  $j = 2$ ,  $K - K^{\leq h} = H_1(U_2 - U_2^{\leq h}) = H_1 P_2 \Psi(q, q^{h+1})$  for integers  $h \geq 2$ , which gives the agreement. Since  $\Psi(q, z) = \sum_{\ell \geq 1} \psi_\ell(q) z^\ell$  converges for small  $|z|$ , locally uniformly in  $q$ , we have  $K(q) - \widehat{K}_h(q) = \sum_{r \geq 1} A_r(q) q^{rh}$  with the  $A_r$  of Appendix D, for all real  $h$  large. Put  $\Phi(\delta, h) = \widehat{K}_h(\rho e^\delta) - 1 = K(\rho e^\delta) - 1 - \sum_{r \geq 1} A_r(\rho e^\delta) \rho^{rh} e^{r h \delta}$ , analytic in  $(\delta, h)$  for  $|\delta|$  small and  $h$  large. On  $|\delta| \leq 1/h^2$ , say,  $r h |\delta| \leq r/h$ , so  $\partial_\delta \Phi = M_\rho + O(h\rho^h) > 0$  and the implicit function theorem gives a unique small root  $\delta(h)$ , analytic in  $h$ , with  $\delta(h) = \alpha \rho^h (1 + O(h\rho^h))$  by the first-order balance  $M_\rho \delta = A_1(\rho) \rho^h + \dots$  and  $A_1(\rho)/M_\rho = \alpha$ . The root is unique on a fixed neighbourhood, not only on  $|\delta| \leq 1/h^2$ : fix  $\delta_0 > 0$  with  $\rho e^{\delta_0} < 1$ . On real  $|\delta| \leq \delta_0$  the tail  $\sum_{r \geq 1} A_r(\rho e^\delta) \rho^{rh} e^{r h \delta}$  and its  $\delta$ -derivative are  $O(h(\rho e^{\delta_0})^h)$ , so  $\Phi(\cdot, h) \rightarrow K(\rho e^\delta) - 1$  in  $C^1$ , uniformly on  $|\delta| \leq \delta_0$ , as  $h \rightarrow \infty$ . The limit vanishes only at  $\delta = 0$  and its derivative  $\rho e^\delta K'(\rho e^\delta)$  is bounded below by a positive constant there, so for large  $h$  the function  $\Phi(\cdot, h)$  is strictly increasing on  $[-\delta_0, \delta_0]$  and  $\delta(h)$  is its only zero in that interval. (These four sentences were added on 5 October 2026 after the independent check recorded in Subsection 21.8.) Next,  $\partial_h \Phi = -\sum_{r \geq 1} r(\log \rho + \delta) A_r(\rho e^\delta) \rho^{rh} e^{r h \delta} = L A_1(\rho) \rho^h (1 + O(h\rho^h)) > 0$ , because  $A_1(\rho) = \rho^2 P_1(\rho) > 0$  and  $L = -\log \rho > 0$ . Hence  $\delta'(h) = -\partial_h \Phi / \partial_\delta \Phi < 0$ .  $\square$

**Theorem 19.2** (Inverse exponential expansion). *Let  $h(\delta)$  be the large real inverse of (19.1), where  $\delta \downarrow 0$ , and put*

$$h_0 = \frac{\log(\alpha/\delta)}{L}. \quad (19.2)$$

Then

$$\boxed{h(\delta) = h_0 + \frac{\delta}{L} \left( h_0 + g - \frac{\rho}{\alpha(1-\rho)} \right) + O(\delta^2 h_0^2)}. \quad (19.3)$$

To every order it has an expansion in powers of  $\delta$  whose coefficients are polynomials in  $h_0$ ; the coefficient at  $\delta^r$  has degree at most  $r$ .

*Proof.* Set  $\epsilon = \rho^h$ . Formula (16.12) is

$$\delta = \alpha\epsilon + \left[ \alpha^2(h + g) - \frac{\alpha\rho}{1 - \rho} \right] \epsilon^2 + O(h^2\epsilon^3).$$

Solving first for  $\epsilon$  and taking its logarithm gives

$$-Lh = \log(\delta/\alpha) - \delta \left( h + g - \frac{\rho}{\alpha(1 - \rho)} \right) + O(\delta^2 h^2).$$

Here  $h = h_0 + O(\delta h_0)$ , so substitution of  $h_0$  in the first correction proves (19.3). Repeating the same substitution with (16.11) gives the asserted polynomial structure at each finite order. The derivative of the logarithmic equation with respect to  $h$  is  $-L + O(\delta h)$ , bounded away from zero, so every formal residual yields a corresponding asymptotic remainder. At order  $N$ , the remainder is  $O_N(\delta^{N+1}h_0^{N+1})$ .  $\square$

At integer cutoffs, a truncated inverse must be accompanied by a rounding interval. The exact interpolation supplies a well-defined crossing; a finite expansion supplies only an approximation to it. Its ceiling is unambiguous only when the approximation is separated from the integers by more than the proved remainder.

## 19.2 Critical counting inversion is almost exactly Lambert- $W$

Let  $L_c = \log(1/\rho_2)$  and  $a_c = \kappa_0/\kappa$ . The entire dominant-pole polynomial in (13.7) has the real interpolation

$$Y_c(x) = \kappa(x + a_c)e^{L_c x}.$$

For sufficiently large  $Y$  its inverse is exactly

$$\boxed{x_c(Y) = \frac{1}{L_c} W_0 \left( \frac{L_c}{\kappa} Y e^{L_c a_c} \right) - a_c.} \quad (19.4)$$

This follows by putting  $z = L_c(x + a_c)$  in  $Y = Y_c(x)$ . No additional inverse-power corrections to this specified interpolant are missing: all errors in replacing the actual coefficients by  $Y_c(n)$  are exponentially smaller.

For example, with  $z_c = (L_c/\kappa)Y e^{L_c a_c}$ ,  $\ell_1 = \log z_c$ , and  $\ell_2 = \log \ell_1$ , ordinary series substitution in  $W + \log W = \log z_c$  gives

$$x_c(Y) = \frac{1}{L_c} \left( \ell_1 - \ell_2 + \frac{\ell_2}{\ell_1} + \frac{\ell_2(\ell_2 - 2)}{2\ell_1^2} + O\left(\frac{\ell_2^3}{\ell_1^3}\right) \right) - a_c. \quad (19.5)$$

The formula (19.4) is preferable numerically; (19.5) exhibits the inverse-logarithmic scale.

**Proposition 19.3** (Discrete crossing bounds). *For  $Y > 0$ , define  $N_c(Y) = \min\{n \geq 1 : f_n(u_c) \geq Y\}$ . For some constant  $C_* > 0$ , with  $R$  as in (13.7),*

$$\left[ x_c(Y) - C_* \frac{R^{-x_c(Y)}}{Y} \right] \leq N_c(Y) \leq \left[ x_c(Y) + C_* \frac{R^{-x_c(Y)}}{Y} \right] \quad (19.6)$$

for all sufficiently large  $Y$ .

*Proof.* Appending a part one creates no new valley and preserves all existing valley weights. It gives a weight-preserving injection from area  $n$  to area  $n + 1$ ; the singleton  $(n + 1)$  is an additional object. Thus  $f_n(u_c)$  is strictly increasing. Near  $x_c(Y)$ ,  $Y'_c(x) \asymp Y$ . The coefficient error  $O(R^{-n})$  therefore moves the real crossing by at most a constant times  $R^{-x_c(Y)}/Y$ . A mean-value bound on a fixed neighborhood of the crossing proves the two inequalities; outside that neighborhood the leading exponential already separates the values. The displayed uncertainty tends to zero because  $R > \rho_2$ .  $\square$

### 19.3 Multicritical counting inverses

At an  $m$ -level critical point, write  $\Pi_m(x) = \kappa^{(m)}x^m(1 + \pi_1/x + \cdots + \pi_m/x^m)$  and  $L_m = \log(1/\rho_{m+1})$  [write: so  $L_1 = L_c$ ]. The leading inverse is

$$x_0(Y) = \frac{m}{L_m} W_0 \left( \frac{L_m}{m} (Y/\kappa^{(m)})^{1/m} \right). \quad (19.7)$$

The inverse of  $e^{L_m x} \Pi_m(x)$  is obtained to all algebraic orders by setting  $x = x_0 + \varphi$  in

$$\varphi = -\frac{1}{L_m} \left[ m \log(1 + \varphi/x_0) + \log \left( 1 + \frac{\pi_1}{x_0 + \varphi} + \cdots + \frac{\pi_m}{(x_0 + \varphi)^m} \right) \right]. \quad (19.8)$$

The right side and its derivative are  $O(x_0^{-1})$ , so local analytic Lagrange inversion is convergent for large  $x_0$  and gives an effective coefficient algorithm. In particular,

$$x = x_0 - \frac{\pi_1}{L_m x_0 + m} + O(x_0^{-2}).$$

The exponentially small coefficient remainder and the same injection give rounding bounds of the form (19.6), with the corresponding polynomial interpolant and radius. These inversion tools are standard; the new input is the explicitly identified critical polynomial and its phase-dependent degree.

### 19.4 Height quantiles and their rounding phase

For fixed  $p \in (0, 1)$ , define the integer quantile  $h_p(n) = \min\{h : \mathbb{P}(\mathbf{H}_n \leq h) \geq p\}$ . The real leading crossing is

$$\beta_p(n) = \frac{\log(\alpha n) - \log(-\log p)}{L}. \quad (19.9)$$

Theorem 17.1 and the nonzero derivative of  $\exp(-\alpha n \rho^x)$  at  $x = \beta_p(n)$  imply, for some  $C_p$ ,

$$\left| \beta_p(n) - C_p \frac{\log n}{n} \right| \leq h_p(n) \leq \left| \beta_p(n) + C_p \frac{\log n}{n} \right|. \quad (19.10)$$

A bare ceiling of (19.9) would omit the narrow but real rounding ambiguity. At criticality, replace the exponential by the monotone mixture Laplace transform in (18.2); its unique positive real crossing in  $\tau$  supplies the analogous quantile constant.

**Proposition 19.4** ([write] Multicritical rounding and critical quantiles). (i) *At the weights (15.1), let  $f_n^{(m)} = [q^n]F_m(q, \mathbf{u}^*)$ ,  $N_m(Y) = \min\{n \geq 1 : f_n^{(m)} \geq Y\}$ , and let  $x_m(Y)$  be the inverse of  $Y_m(x) = e^{L_m x} \Pi_m(x)$  on its eventually increasing branch. With  $R$  as in (15.4), there is  $C_* > 0$  such that, for all sufficiently large  $Y$ ,*

$$\left\lfloor x_m(Y) - C_* \frac{R^{-x_m(Y)}}{Y} \right\rfloor \leq N_m(Y) \leq \left\lceil x_m(Y) + C_* \frac{R^{-x_m(Y)}}{Y} \right\rceil.$$

(ii) In the setting of Theorem 18.1 with fixed  $s$ , let  $\Phi_s(\tau)$  be the main term of (18.2) as a function of  $\tau = \tau_n$ . For fixed  $p \in (0, 1)$  it has a unique root  $\tau_p > 0$  of  $\Phi_s(\tau) = p$ ; put  $\beta_p^{\text{crit}}(n) = (\log n - \log \tau_p)/L_m$ . Then  $h_p(n) = \min\{h : \mathbb{P}_{n, u_n}(\mathbf{H}_n \leq h) \geq p\}$  satisfies, for some  $C_p$  and all large  $n$ ,

$$\left\lceil \beta_p^{\text{crit}}(n) - C_p \frac{\log n}{n} \right\rceil \leq h_p(n) \leq \left\lceil \beta_p^{\text{crit}}(n) + C_p \frac{\log n}{n} \right\rceil.$$

The same argument, with  $\Phi(\tau) = e^{-\alpha\tau}$  and Theorem 17.1, proves (19.10).

*Proof.* This proof was added in writing; the manuscript states (i) and (ii) by analogy in the two paragraphs above. (i) Appending a part one to a valley-monotone composition creates no valley (the new last run is an endpoint run, and the run before it gains a smaller right neighbour, so it is not a valley) and removes none, so it is a weight-preserving injection from area  $n$  to area  $n+1$ ; the singleton  $(n+1)$ , of weight one, is not in its image. Hence  $f_{n+1}^{(m)} > f_n^{(m)}$ . By (15.4),  $f_n^{(m)} = Y_m(n) + O(R^{-n})$ , and  $Y'_m(x) = Y_m(x)(L_m + \Pi'_m(x)/\Pi_m(x)) \asymp Y_m(x)$  for large  $x$ . The proof of Proposition 19.3 now applies verbatim with  $Y_m$ ,  $x_m$  in place of  $Y_c$ ,  $x_c$ .

(ii) With  $S = \sum_i \alpha_i^* \xi_i$ ,  $\Phi_s(\tau)$  is the expectation of  $e^{-\tau S}$  under the tilted law  $\pi_s$  of Proposition 18.2. Since  $S \geq \min_i \alpha_i^* > 0$ ,  $\Phi_s$  is continuous,  $\Phi_s(0) = 1$ ,  $\Phi_s(\tau) \rightarrow 0$  as  $\tau \rightarrow \infty$ , and  $\Phi'_s(\tau) = -\mathbb{E}_{\pi_s}[S e^{-\tau S}] < 0$ ; so  $\tau_p$  exists and is unique. Put  $g(x) = \Phi_s(n\rho_{m+1}^x)$  for real  $x$ , so  $g(\beta_p^{\text{crit}}(n)) = p$  and  $g'(x) = -L_m \tau \Phi'_s(\tau)$  at  $\tau = n\rho_{m+1}^x$ , which is  $\geq c > 0$  while  $\tau \in [\rho_{m+1}^2 \tau_p, \tau_p / \rho_{m+1}^2]$ . Apply Theorem 18.1 with  $\tau_- = \rho_{m+1}^2 \tau_p$ ,  $\tau_+ = \tau_p / \rho_{m+1}^2$ ; its error is at most  $K \log n / n$  there. Let  $\varepsilon_n = C_p \log n / n$  with  $C_p > K/c$ . The integer  $h^+ = \lceil \beta_p^{\text{crit}} + \varepsilon_n \rceil$  has  $\tau$  in that range and  $g(h^+) \geq p + c\varepsilon_n$ , so  $\mathbb{P}(\mathbf{H}_n \leq h^+) \geq p + c\varepsilon_n - K \log n / n > p$  and  $h_p(n) \leq h^+$ . The integer  $h^- = \lceil \beta_p^{\text{crit}} - \varepsilon_n \rceil - 1$  lies in  $[\beta_p^{\text{crit}} - \varepsilon_n - 1, \beta_p^{\text{crit}} - \varepsilon_n]$ , so its  $\tau$  is in range and  $g(h^-) < p - c\varepsilon_n$ ; hence  $\mathbb{P}(\mathbf{H}_n \leq h) \leq \mathbb{P}(\mathbf{H}_n \leq h^-) < p$  for every  $h \leq h^-$ , and  $h_p(n) \geq h^- + 1$ . These are the two ceilings.  $\square$

[write] **Added 5 October 2026, batch 98: instances of repository results.** As for Part I's inverse (the notes in Sections 7 and 8), the inversion tools of this section are instances of the transseries-and-inversion volume [5], and the manuscript itself calls them standard. Writing  $X = x + a_c$ ,  $Y = Y_c(x)$  reads  $L_c X + \log X = \log(Y/\kappa) + L_c a_c$ , the Lambert core `p0:thm:lambert-core` with  $a = L_c$ ,  $b = 1 > 0$  (principal branch); this is (19.4), and (19.5) is its standard logarithmic series. At  $m$  levels the leading block  $L_m x + m \log x = \log(Y/\kappa^{(m)})$  is the same core with  $a = L_m$ ,  $b = m$ , giving (19.7), and (19.8) is reversion about that core (`p0:thm:core-reversion`), i.e. Lagrange inversion (`p0:thm:LB`), which also underlies Theorem 19.2. The reduced equation  $\delta = \alpha e e^{h\delta}$  in the proof of Theorem 16.2 is solved by the Lambert function in the same way. Propositions 19.3 and 19.4(i), and (19.10), are the staircase theorem `p0:thm:staircase`, parts (1)–(2), whose generic arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; those statements concern an arbitrary monotone interpolation and give no formal status to the propositions here. No novelty is claimed for the methods; the constants and the critical polynomials are specific to this report.

## 20 Reproducible checks and numerical behavior

### 20.1 Exact finite verification

The supplied `code/verify.py` uses only Python's standard library. It computes coefficients by triangular formal-series division over the integers. Independently it generates every composition through area fourteen, extracts plateau valleys directly, and compares six weighted or height-restricted settings. All 84 comparisons pass. The settings include unweighted objects, maximum

heights three and five, weight two at level one, and simultaneous weights two and three at levels one and two. It also checks A001523 through area twenty and retains unweighted coefficients through area 240.

The exact denominator for a marked level is

$$Q_{j,u}(q) = (1 - q^j)^2 - uq^j(1 - q^j)U_{j+1}(q), \quad (20.1)$$

and the numerator in  $Q_{j,u}D_j$  is

$$q^j(1 - q^j) - uq^{2j}U_{j+1}(q) + D_{j+1}(q). \quad (20.2)$$

Because  $Q_{j,u}(0) = 1$ , no division of integers is needed. At height cap  $h$ , initialize  $U_{h+1} = D_{h+1} = 0$ ; without a cap, initializing at level  $N + 1$  is exact through degree  $N$ . These are finite identities, not asymptotic approximations.

[write] **Added 5 October 2026, batch 98.**  $Q_{j,u} = (1 - q^j)(1 - q^j - uq^jU_{j+1})$ , so  $Q_{j,1} = (1 - q^j)Q_j$  with Part I's  $Q_j$  of (2.2), and (20.1)–(20.2) at  $u = 1$  are the last line of (A.1).

## 20.2 Critical coefficients and the double pole

The numerical programs use 65-digit `mpmath` arithmetic with a maximum level of 320. Comparing levels 220 and 320 in  $K(\rho_2)$  gives a difference about  $1.43 \cdot 10^{-43}$ . This is a convergence diagnostic, not a rigorous interval certificate. The exact critical constants are always the analytic quantities in (13.2)–(13.8).

| $n$   | $\rho_2^n f_n(u_c) - (\kappa n + \kappa_0)$ | $\mathbb{E}V_n/n$ | $\text{Var } V_n/n^2$ |
|-------|---------------------------------------------|-------------------|-----------------------|
| 30    | $4.6043 \cdot 10^{-2}$                      | 0.02736564        | 0.00084894            |
| 60    | $2.2957 \cdot 10^{-3}$                      | 0.03149433        | 0.00065425            |
| 120   | $5.5586 \cdot 10^{-6}$                      | 0.03413759        | 0.00055328            |
| 240   | $3.2472 \cdot 10^{-11}$                     | 0.03564653        | 0.00050604            |
| Limit | 0                                           | 0.03728019        | 0.00046327            |

The exponentially small residual concerns the coefficient formula. The moments need not converge exponentially: marking splits the double pole, and their leading finite-size corrections are algebraic.

At  $n = 240$  the leading critical-window formula, its first correction, and the independently computed scaled coefficient are:

| $s$ | $\kappa n G(s/M_c)$ | with $G_1(s)$ | $\rho_2^n f_n(u_c e^{s/n})$ |
|-----|---------------------|---------------|-----------------------------|
| −4  | 48.27503236         | 48.32923935   | 48.31748024                 |
| 0   | 55.83114319         | 55.49838473   | 55.49838473                 |
| 4   | 65.04998792         | 64.24315456   | 64.26440169                 |

The remaining error for nonzero  $s$  is consistent with the proved  $O(1/n)$  scaled remainder; the table is not used to infer that bound.

## 20.3 Extreme heights and pole shifts

For the ordinary model, the exact finite coefficient ratio and the leading height prediction in (17.1) give:

| $n$ | $h$ | exact $\mathbb{P}(H_n \leq h)$ | $e^{-\alpha n \rho^h}$ |
|-----|-----|--------------------------------|------------------------|
| 60  | 5   | 0.39227262                     | 0.40832225             |
| 120 | 6   | 0.39622717                     | 0.40398216             |
| 240 | 7   | 0.39582494                     | 0.39963725             |

The visible finite-size differences are retained rather than hidden by a fitted intensity. The pole-shift expansion itself converges much faster:

| $h$ | relative error, first term | relative error, two terms |
|-----|----------------------------|---------------------------|
| 8   | $-1.7175 \cdot 10^{-2}$    | $-4.3993 \cdot 10^{-4}$   |
| 12  | $-1.6241 \cdot 10^{-3}$    | $-3.9517 \cdot 10^{-6}$   |
| 16  | $-1.3958 \cdot 10^{-4}$    | $-2.9209 \cdot 10^{-8}$   |
| 24  | $-8.8483 \cdot 10^{-7}$    | $-1.1741 \cdot 10^{-12}$  |
| 32  | $-5.0260 \cdot 10^{-9}$    | $-3.7886 \cdot 10^{-17}$  |

Here “relative error” means approximation divided by the computed  $\delta_h$ , minus one.

At the first critical point, the numerically evaluated finite weighted coefficient ratios and the mixture prediction (18.5) include

| $n$ | $h$ | computed critical CDF | mixture prediction |
|-----|-----|-----------------------|--------------------|
| 120 | 7   | 0.19320378            | 0.20584639         |
| 120 | 8   | 0.37460867            | 0.36990049         |
| 240 | 9   | 0.28593991            | 0.28662767         |
| 240 | 10  | 0.46399389            | 0.45564238         |

These computations use the explicit weighted recurrence, not samples from a Monte Carlo generator.

Finally, at the two-level critical point,

$$\rho_3 = 0.6949152510\dots, \quad u_1^* = 0.02240337599\dots, \quad u_2^* = 0.21458477859\dots$$

For  $n = 240$  and  $(s_1, s_2) = (3, -2)$ , the numerically evaluated coefficient ratio with the  $1/n$  weight tilts is  $1.0118132157\dots$ , whereas the limiting simplex integral is  $1.0129148706\dots$ . For  $(-3, 2)$  the corresponding numbers are  $0.9911396048\dots$  and  $0.9897390369\dots$ . These are additional finite checks of the multilevel construction, not substitutes for Theorem 15.2.

## 20.4 Running the package

From the package directory:

```
python code/verify.py
python code/symbolic_checks.py
python code/numerics.py
python code/critical_extremes.py
pdflatex article.tex
pdflatex article.tex
```

The first script requires only Python’s standard library; symbolic checks require `sympy`, and numerical diagnostics require `mpmath`. The data directory records the outputs. No external

papers, repository source files, or font files are included. The scripts do not change the remote repository or submit anything to OEIS.

[write] **Added 5 October 2026, batch 98.** The commands above describe the delivered package, not this repository. The four programs are shipped in `code/` and their outputs in `data/`, each under its delivered name with the prefix `02-critical-windows-` (for example `code/02-critical-windows-verify.py`); the delivered article, its PDF and its build script are not shipped (this report’s README gives the build command). The programs write unprefixed files into a sibling `data/` directory, `numerics.py` and `critical_extremes.py` import `verify` and `numerics` by their delivered names (inside the repository those names would resolve to Part I’s programs, which lack the imported functions), and `critical_extremes.py` reads `numerical_checks.json`. The README therefore reruns them on a scratch copy with the delivered names restored. The rerun of the write (5 October 2026, Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0) passed: 84 enumeration checks and A001523 through area 20, five symbolic identity groups, and the two numerical programs, all four outputs equal to the shipped files as JSON (the bytes differ only by CRLF line ends written on Windows). In the data keys, `R` is  $\rho_2$ , `R3` is  $\rho_3$ , `P` is  $P_1(\rho)$ , `A` is  $\alpha$ , `physical_M`, `physical_V` are Part I’s  $M$ ,  $V$ , `critical_M`, `critical_V`, `Jc`, `critical_D`, `critical_d` are  $M_c$ ,  $V_c$ ,  $J_c$ ,  $D_c$ ,  $d_c$ , `critical_height_alpha1` and `critical_height_alpha2` are  $\alpha_1^*$  and  $\alpha_0^*$ , `critical_level2_M` is  $M_2(\rho_2)$ , `shift` is  $a_c$ , and `u1`, `u2`, `M1`, `M2` of the two-level record are  $u_1^*$ ,  $u_2^*$ ,  $M_1^*$ ,  $M_2^*$  at  $m = 2$ . Keys such as `exact_cdf` and `exact_ratio` denote the finite recurrence evaluated at approximate weights, not enclosures. The A001523 terms through area 20 hard-coded in `02-critical-windows-verify.py` are OEIS data (CC BY-SA 4.0); the arrays of  $D_1$ ,  $U_2$ ,  $D_2$  in `02-critical-windows-exact_checks.json` are computed, not OEIS data.

## 21 Further questions and research

[write] **Added 5 October 2026, batch 98.** Subsections 21.1–21.7 are the manuscript’s Section 11 (“Further research questions”), printed with its numbering of topics 1–7. Subsection 21.8 lists, under the repository’s standing rule of 4 October 2026, every statement of the manuscript that it supports only by a sketch, an analogy or a one-line remark, with what the write did about each, and the part of Part I’s question that Part II leaves open.

### 21.1 A critical extremal point process

Theorem 18.1 identifies the law of the largest column. A stronger next target is the joint distribution of the largest several column heights, with their plateau multiplicities and the levels of their renewal components. A natural conjecture is a component-marked Poisson limit conditional on the Dirichlet fractions, with a corresponding random-intensity mixture after conditioning is removed. The current maximum-height proof alone does not establish such a point-process limit: multiple columns in one unimodal unit can be dependent and must be controlled explicitly.

### 21.2 The first correction to the mixed extreme law

The present error in (18.2) is  $O(\log n/n)$ . Derive the full correction, including the perturbed residues, the next kernel-tail term, and the simplex boundary terms in the coefficient sum. The exact  $\Psi$ -expansion and the cutoff polynomials show that the correction should involve the lattice phase and powers of  $\log n$ . This would sharpen critical height quantiles beyond the current rounding band.

### 21.3 Growing numbers of critical levels

All multicritical theorems here keep  $m$  fixed. Determine how fast  $m = m(n)$  may grow while the Dirichlet approximation remains uniform. Since  $\rho_{m+1} \rightarrow 1$ , analytic neighborhoods shrink and the number of possible competing poles increases. Neither the fixed- $m$  simplex proof nor the fixed-minimum-part asymptotic theorem justifies this interchange of limits.

### 21.4 A local limit theorem for coexistence fractions

The bounded-support weak limits proved here determine macroscopic moments. A quantitative joint local limit for  $(V_{1,n}, \dots, V_{m,n})$  would describe individual lattice probabilities, including faces of the simplex. Interior Gaussian fluctuations around a prescribed Dirichlet fraction should interact with boundary layers where one renewal family has only finitely many units. A uniform theorem across those layers remains to be proved.

### 21.5 Nonreal poles and optimal exponential errors

The proofs require a radius slightly beyond the dominant singularity, not a certification of the next singularity. Locate and certify the first several nonreal poles of the weighted and height-capped models. This would make the exponential remainder effective, identify oscillatory correction sectors, and improve the discrete counting inverses near difficult thresholds.

### 21.6 Natural boundaries and moving minimum parts

The unit-circle natural-boundary question and the moving-lower-bound crossover problem in Part I remain separate questions. The present  $q$ -product identity may help study root-of-unity behavior, but it does not prove that singularities are noncancelling near every boundary point. Likewise, increasing height cutoff and increasing minimum part are different limits; the maximum-height theorem does not solve the latter crossover problem.

[write] **Added 5 October 2026, batch 98.** These are Part I's questions "Is the unit circle a natural boundary?" and "Moving lower bounds and a crossover problem" (Section 10), restated; Part II answers neither.

### 21.7 Formalization and effective certificates

A first formalization target is the unique marked decomposition, including plateau semantics and height caps. Next come exact truncated-series identities and the elementary simplex coefficient lemma. Certified interval bounds for  $u_c$ ,  $\alpha$ , and the critical intensities would be useful independently of a full complex-analytic formalization. The numerical agreement in this package is not a replacement for those certificates.

### 21.8 Sketched claims of the manuscript, and what remains of Part I's question

[write] **Added 5 October 2026, batch 98.** Each item names the manuscript's claim, its support there, and its status now. The proofs supplied in writing have not been independently reviewed; until they are, the items they settle should be read as settled only to that standard.

1. *Conditional intensity of the critical maximum* (manuscript Section 8.1; here the paragraph before Proposition 18.2). Claim: conditionally on the limiting area fractions  $X_i$ , the extreme-height law

- has intensity  $\sum \alpha_i^* X_i$ , so the critical maximum is generally not asymptotically independent of the fractions. Support: one sentence (“Joint marking of the area fractions shows more”). Missing: the simplex lemma with simultaneous area marks and the cutoff, and the identification of the joint limit. Status: proved in writing, Proposition 18.2, which also makes “generally” exact (independence holds if and only if all  $\alpha_i^*$  coincide).
2. *Tilted multicritical window* (manuscript Section 5.2). Claim: the weights  $u_j^* e^{s_j/n}$  multiply the leading coefficient by  $\mathcal{G}_m(\mathbf{s})$  and tilt the Dirichlet law. Support: “The proof is the same nearby-pole argument.” Status: proved in writing, Proposition 15.4.
  3. *Complete fixed-parameter classification* (manuscript Section 5.2). Claim: with  $\nu$  minimal roots, the pole has order  $\nu$  and the coefficients are  $r_*^{-\nu}$  times a polynomial of degree  $\nu - 1$ . Support: a paragraph sketch (noncancellation through the suffix after the last active level). Status: proved in writing, Proposition 15.5.
  4. *Multicritical rounding and critical quantiles* (manuscript Sections 9.3 and 9.4). Claims: rounding bounds “of the form” (19.6) at every multicritical point, and the critical analogue of (19.10) through the unique crossing of the mixture transform. Support: analogy. Status: proved in writing, Proposition 19.4.
  5. *Monotonicity of the interpolated cutoff shift* (manuscript Section 9.1). Claim:  $\delta(h)$  is strictly decreasing for large real  $h$ . Support: one line. Status: proved in writing, Lemma 19.1.
  6. *The residue ratio* (17.4) in the proof of Theorem 17.1: argued in outline; recorded as complete in outline, not as an open question.
  7. *Part I’s question “Joint limits and extreme heights”* (Section 10): its extreme-height half is answered (Theorems 17.1, 17.2). Its other half stays open and is not addressed by the manuscript: a joint normal law for the additive statistics that the renewal representation can mark (number of columns, number of ones, peak heights), which needs a proof that the relevant covariance matrix is nondegenerate.

No claim of the manuscript was found to be wrong.

*Dated note (5 October 2026).* An independent adversarial check of the results added at this write, made by the intake after the write, found all of them valid: Propositions 15.4, 15.5, 18.2 (both directions of its “if and only if”) and 19.4, Lemma 19.1, and the outline of the residue ratio (17.4), whose factor  $h$  in  $O(h\rho^h)$  is genuinely present. The check was a careful reading with independent computations: a new power-series implementation of the level decomposition with height cutoffs and area marks, validated against brute-force enumeration of compositions up to area 14; the constants  $\rho$ ,  $\rho_2$ ,  $\rho_3$ ,  $u_c$ ,  $\alpha$ ,  $\alpha_0^*$ ,  $\alpha_1^*$  and  $M_c$  recomputed to 40 digits; and exact-coefficient tests at  $m = 1$  and  $m = 2$  of the tilted window, of every chamber type, of the conditional law against the independence alternative (the first has error  $O(\log n/n)$ , the second a fixed gap), and of the rounding and quantile brackets. It is not a formal verification or an external review. One fix was made: the proof of Lemma 19.1 established the uniqueness of the root only on  $|\delta| \leq 1/h^2$ , while the statement says “with  $\delta$  small”; the proof now also gives uniqueness on a fixed neighbourhood  $|\delta| \leq \delta_0$  with  $\rho e^{\delta_0} < 1$ , because the equation tends uniformly to  $K(\rho e^{\delta}) - 1$ . The check also observed, numerically only, that  $\hat{K}_h = K^{\leq h}$  holds at  $h = 1$  as well, so the restriction to integers  $h \geq 2$  in the lemma, which comes from the range  $h \geq j$  of (16.1), is more cautious than needed.

## 22 Claim ledger and conclusion of Part II

| Statement                                             | Status and boundary                                                                                      |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| A001523 formulas and classical asymptotics            | Prior literature and OEIS; not claimed as new. Only the block connection is used.                        |
| Minimum-part decomposition and unweighted simple pole | Prior results, reproved here to establish the hypotheses.                                                |
| Bottom-valley phase transition                        | Proved for fixed positive weight; exact critical value, three regimes, and critical window are explicit. |

| Statement                         | Status and boundary (continued)                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| All window orders                 | Finite-order, uniform additive expansions, with an algorithm from local Taylor and Laurent data.                                             |
| Multicritical Dirichlet law       | Proved for any fixed finite number of marked levels; no growing-level uniformity is asserted.                                                |
| Ordinary maximum-height law       | Proves the lattice-Gumbel question posed in the previous report, with explicit intensity and $O(\log n/n)$ window error.                     |
| Critical maximum-height law       | Proved Dirichlet-mixture formula, including simultaneous $1/n$ weight tilts and a uniform height-window error.                               |
| Cutoff-pole expansion and inverse | All finite exponential orders with polynomial coefficients and explicit remainders; no unrestricted convergence or Stokes claim.             |
| Discrete inverses and quantiles   | Specified smooth interpolants plus two ceilings; no unconditional rounding shortcut.                                                         |
| Exact tests and numerical data    | Independent finite integer checks pass; decimal values are reproducible diagnostics, not interval-certified constants.                       |
| Novelty and formal status         | An extension of documented repository work; no exhaustive priority claim, no Lean verification, and no independent peer review are asserted. |

[write] **Added 5 October 2026, batch 98.** “The previous report” is Part I. The row “Cutoff-pole expansion and inverse” was delivered as “Cutoff-pole transseries and inverse” (see the note after Theorem 16.2); the table is set ragged-right. The statements completed in writing are listed in Subsection 21.8.

The central conclusion is that the same OEIS-linked renewal decomposition supports two different kinds of randomness. In the ordinary unweighted regime, one renewal family occupies almost all the area, its valley count is Gaussian, and its tallest column obeys a lattice-Gumbel law independent of that count. At coexistence, several families share macroscopic area with a Dirichlet law; their joint extreme distribution retains that random allocation as a mixture intensity. The pole multiplicity controls the counting polynomial, the critical-window scale, the macroscopic covariance, and the appropriate Lambert- $W$  inverse. The exact height-tail identity makes the exponentially small cutoff corrections computable rather than merely qualitative.

## A A coefficient recurrence suitable for exact checking

Let all series be truncated modulo  $q^{N+1}$ . For descending  $j = N, N-1, \dots, 1$ , compute

$$\begin{aligned} Q_j &= 1 - q^j - q^j U_{j+1}, \\ (1 - q^j)^2 U_j &= q^j (1 - q^j) + U_{j+1}, \\ (1 - q^j) Q_j D_j &= q^j Q_j + D_{j+1}. \end{aligned} \tag{A.1}$$

Every denominator has constant term one. If  $v(q)w(q) = b(q)$ ,  $v(0) = 1$ , its coefficients are obtained by

$$w_n = b_n - \sum_{k=1}^n v_k w_{n-k}.$$

This requires no division in the coefficient ring. Initialization with zero at level  $N+1$  is exact at the requested order, so integrality of every computed coefficient is automatic. The enumeration proof in Section 2, not integrality alone, identifies the objects being counted.

## B Claim ledger and limitations

| Claim                                     | Status and justification                                                                                                                        |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Published generating function             | Prior result of [1]; independently rederived by the unique minimum-part decomposition.                                                          |
| Pure exponential equivalent               | Proved in Theorem 4.1, with exact constants and an exponentially smaller error.                                                                 |
| Corrected numerical constants             | Certified by standard-library integer interval arithmetic with explicit infinite-tail bounds.                                                   |
| Infinite positive poles; non-D-finiteness | Proved without a noncancellation assumption in Theorem 3.4 and Corollary 3.5.                                                                   |
| Arbitrary exponential cutoff              | Proved in Theorem 5.1; not a claim of convergence of an infinite pole sum.                                                                      |
| Limit laws                                | Proved by uniform analytic perturbation and direct characteristic-function convergence.                                                         |
| All logarithmic pole orders               | Proved in Theorem 7.3; the $1/j$ -scale correction remains to be sharpened.                                                                     |
| Inverse expansion                         | Proved for a specified finite-sector interpolant; integer crossings are bounded with two ceilings.                                              |
| Novelty and formal status                 | No matching prior proof found in the targeted search. No exhaustive priority claim, no Lean proof, and no independent peer review are asserted. |

## C Computing the Laurent constants without unstable subtraction

At  $\rho_2$ , write

$$D_2(q) = \frac{c_2}{1 - q/\rho_2} + c_{2,0} + O(1 - q/\rho_2).$$

Let

$$B_2(q) = E_2(q)D_3(q), \quad M_2(\rho_2) = \rho_2 K_2'(\rho_2), \quad J_2 = \rho_2^2 K_2''(\rho_2)/2.$$

Since  $K_2(\rho_2) = 1$ , direct expansion of (12.3) gives

$$c_2 = B_2(\rho_2)/M_2(\rho_2), \tag{C.1}$$

$$c_{2,0} = H_2(\rho_2) + \frac{B_2(\rho_2)J_2}{M_2(\rho_2)^2} - \frac{\rho_2 B_2'(\rho_2)}{M_2(\rho_2)}, \tag{C.2}$$

$$b = \frac{c_2}{(1 - \rho_2)^2}, \tag{C.3}$$

$$b_0 = \frac{c_{2,0}}{(1 - \rho_2)^2} - \frac{2\rho_2 c_2}{(1 - \rho_2)^3}. \tag{C.4}$$

These formulas avoid subtracting two large nearby singular values of  $B$ . They are the formulas used in the numerical program.

[write] **Added 5 October 2026, batch 98.** Here  $c_2$  is the dominant amplitude of  $D_2$  (as in Theorem 4.1), not the amplitude  $C_2$  of  $D_1$  at  $\rho_2$ ; the manuscript's  $A_2$ ,  $d_{20}$  and  $M_2 = RK_2'(R)$  are printed as  $B_2$  ( $B_2 = E_2 D_3$  is the level-2 analogue of  $B = E_1 D_2$ ),  $c_{2,0}$  and  $M_2(\rho_2)$ .

## D A fully finite prescription for all height-cutoff coefficients

For a requested order  $N$ , compute  $\gamma_0(q), \dots, \gamma_{N-1}(q)$  by (16.7) and then the first  $N$  coefficients of  $\Psi$  by (16.6). Put

$$A_r(q) = H_1(q)P_2(q) [z^r] \Psi(q, z) q^r.$$

These satisfy  $K(q) - K^{\leq h}(q) = \sum_{r \geq 1} A_r(q) q^{rh}$ . Now introduce a formal indeterminate  $\epsilon$  and polynomials  $\varpi_0(h), \dots, \varpi_{N-1}(h)$ . In

$$K \left( \rho \exp \left( \sum_{r=1}^N \varpi_{r-1}(h) \epsilon^r \right) \right) - 1 = \sum_{r=1}^N A_r \left( \rho \exp \left( \sum_{\ell=1}^N \varpi_{\ell-1}(h) \epsilon^\ell \right) \right) \epsilon^r \exp \left( rh \sum_{\ell=1}^N \varpi_{\ell-1}(h) \epsilon^\ell \right),$$

compare coefficients through  $\epsilon^N$ . At step  $r$  divide the known residual by  $M_\rho$  to obtain  $\varpi_{r-1}$ . Only derivatives of analytic functions at one ordinary point  $\rho < 1$  occur. Theorem 16.2 proves that this finite symbolic prescription has the asserted asymptotic meaning.

## References

- [1] R. Flórez, J. L. Ramírez, and D. Villamizar, *Restricted bargraphs and unimodal compositions*, Journal of Combinatorial Theory, Series A **208** (2024), 105934. doi:10.1016/j.jcta.2024.105934. Especially Section 2 and Conjecture 2.3. Publisher HTML consulted October 1, 2026.
- [2] OEIS Foundation Inc., *A001523: Number of stacks, or planar partitions of  $n$ ; also number of weakly unimodal compositions of  $n$* . <https://oeis.org/A001523>. Entry consulted October 1, 2026. The entry credits Auluck's known leading asymptotic and links the bargraph paper above.

- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Chapters IV–V for rational and meromorphic asymptotics, and Chapter IX for parameter distributions. Author’s book page:  
<https://sedgewick.io/books/analytic-combinatorics/>.
- [4] V. Reshetnikov, *ProveIt*, project repository, particularly `Analysis/Transseries/README.md`.  
<https://github.com/VladimirReshetnikov/ProveIt>. Read-only inspection, October 1, 2026. No repository changes were made.
- [5] *Transseries and inversion*, a volume of the ProveIt repository:  
`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (theorems `p0:thm:lambert-core`, `p0:thm:core-reversion`, `p0:thm:staircase` and `p0:thm:LB`). Added in writing, batch 75.